

A Probabilistic Gaia DR3 Catalogue of Stars with Very Low Galactic Rest-Frame Speeds

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ABSTRACT

Context. A star observed at very low Galactic rest-frame speed is caught close to an orbital turning point, and its specific angular momentum is bounded above by the product of its Galactocentric radius and its observed speed. In a static, centrally concentrated potential its apocentre therefore lies at or just beyond its present radius, making the low- V_{grf} state a kinematic selection with a largely predictable orbital corollary. Gaia-era catalogues selected at the opposite, high-speed extreme are numerous, whereas the slow tail has not previously been surveyed systematically on total rest-frame speed.

Aims. We aim to construct and publicly release a probabilistic Gaia DR3 catalogue of stars consistent with the low- V_{grf} condition and to separate the model-robust from the model-sensitive components of its orbital characterisation.

Methods. Monte Carlo propagation of the full Gaia parallax–proper-motion covariance, with Lindegren et al. (2021) parallax zero-points and Bailer-Jones et al. (2021) photogeometric distances, yields per-star membership probabilities for $V_{\text{grf}} < 25 \text{ km s}^{-1}$. Orbits are integrated in the axisymmetrized Hunter et al. (2024) potential and stress-tested against barred, halo-mass, second-potential, and Solar-parameter variants; observability-matched Gaia control samples at higher V_{grf} provide an empirical null.

Results. Membership probabilities are computed both as forward Monte Carlo scores and, because the parent population rises steeply with V_{grf} , after reconstructing the latent V_{grf} distribution; the latter correction lowers the score-implied purity of a forward-defined $P > 0.50$ sample from 73% to 42%, and we therefore define the catalogue on the population-prior probability. It comprises a primary Tier A+B sample of 276 stars ($P > 0.84$, purity 95%) within a broader Tier A+B+C sample of 621 ($P > 0.50$, 79%); these are strict nested subsets of the 541 and 1,952 stars selected by the forward scores, which are also released. The sample is strongly radial, with compact cylindrical pericentres (Monte Carlo posterior-median eccentricity 0.965, median $R_{\text{peri}} = 153 \text{ pc}$ and $R_{\text{apo}} = 8.24 \text{ kpc}$), distinctly more so than the matched controls. The matched comparison is used as a null check that the slow state is not a generic property of Gaia 6D stars, not as a calibrated speed–eccentricity relation.

Conclusions. We release the tiered catalogue, the per-star Monte Carlo and orbit products, and subset-qualified chemistry; completeness is set by the sparse Gaia RVS selection function, not by the velocity cut.

Key words. catalogs – astrometry – stars: kinematics and dynamics – Galaxy: kinematics and dynamics – Galaxy: structure – methods: statistical

1. Introduction

Gaia DR3 delivers full six-dimensional phase-space coordinates for tens of millions of stars by combining its astrometry with radial velocities from the RVS instrument (Gaia Collaboration et al. 2023; Katz et al. 2023). Most Gaia-era Galactic archaeology uses that information the same way: identify a population—a disc component, a halo substructure, a stream of accreted debris—and then characterise its kinematics (see Binney & Merrifield 1998; Bland-Hawthorn & Gerhard 2016; Helmi 2020; Deason & Belokurov 2024, for reviews). The same data support a complementary approach: select directly on a rare kinematic state, and ask afterwards what kinds of star occupy it. This paper takes that approach for one such state, stars observed with unusually low instantaneous speed in the Galactic rest frame (V_{grf}).

Selection at a kinematic extreme is already well established at the fast end. Gaia DR3 has been searched for hypervelocity and high-velocity candidates (Marchetti et al. 2022; Carballo-Bello et al. 2025) and for halo samples defined by high tangential velocity or reduced proper motion (Viswanathan et al. 2023), and the fast tail has long been used to

constrain the Galactic escape speed and the mass of the Milky Way (Smith et al. 2007; Piffl et al. 2014; Koppelman & Helmi 2021). The slow end has attracted much less systematic attention. The closest Gaia-era counterpart to the present work is Filion et al. (2025), who identify ~ 70 stars with low or retrograde Galactocentric azimuthal velocity and thin-disc-like abundances. That selection is made on a single velocity component within a chemical box; ours is made on the full rest-frame speed and applies no chemical condition, so the two samples are complementary rather than competing—we compare them directly in Appendix F. We are not aware of an existing Gaia-era catalogue selected on total rest-frame speed at this extreme, and assembling one is the purpose of this paper.

A related motivation comes from empirical stellar-tracer approaches to the local dark-matter velocity distribution, which use metal-poor and accreted populations as observable proxies for the accreted dark matter (Herzog-Arbeitman et al. 2018a,b; Necib et al. 2019a,b; Evans et al. 2019); the low-speed part of that distribution is of particular interest because solar gravitational focusing acts most strongly on slow particles (Lee et al. 2014). We make no dark-matter inference here, and our selection function is set by Gaia RVS rather than by any

halo-tracer criterion, so the catalogue is context for that literature rather than a constraint on it.

The low-speed state is worth isolating because it carries a dynamical implication that a single epoch of data can support. A star observed almost at rest with respect to the Galactic centre is, at that instant, close to a turning point of its orbit: in a static, centrally concentrated potential its apocentric radius lies at or just beyond its present Galactocentric radius. Its specific angular momentum is likewise bounded above by the product of its Galactocentric radius and its observed speed, which for a star near the solar radius moving at $< 25 \text{ km s}^{-1}$ is under 11% of the circular-orbit value at the same radius. Both statements follow from the observed velocity vector alone and are insensitive to the choice of potential.

What the low speed does *not* fix is the depth of the inner turning point. Because cylindrical pericentre is controlled by the small residual angular momentum, orbits consistent with the same low speed span orders of magnitude in cylindrical pericentric distance—from the kiloparsec-scale pericentres typical of radial accretion debris in the apocentre-pile-up picture (Deason et al. 2018) to very small cylindrical pericentres for the lowest angular momenta. A small cylindrical pericentre by itself establishes only proximity to the Galactic spin axis; a parsec-scale passage near Sgr A* additionally requires small $|z|$, and is assessed separately below using the three-dimensional closest approach. In either case the star crosses the inner Galaxy quickly and spends most of each radial period near apocentre; what distinguishes the configurations is how closely the orbit approaches the Galactic centre. Quantifying that closest approach star by star, from the full velocity vector and its uncertainties, is what the orbit modelling below sets out to do.

It is worth being explicit about the geometry this implies, because a small pericentre is easy to picture as a comet-like turning point at which the star reverses and retraces its path. That is not what these orbits do. For a closed Kepler ellipse the azimuth swept between successive apocentres is 360° , so the star does return along its inbound track; for a harmonic, constant-density core it is 180° , so successive apocentres lie on opposite sides of the centre. In the adopted potential the catalogue sits close to the latter: the median azimuth swept per radial period is 195° (16th–84th percentile 155° – 202°), and no catalogue member exceeds 300° (Section 4). These stars therefore transit the inner Galaxy and emerge on the far side, rather than reaching inward and turning back.

The sample is therefore defined by present-day kinematics, not by inferred origin. We construct a probabilistic low- V_{grf} Gaia DR3 catalogue, quantify how stable velocity-threshold membership is under measurement uncertainties, and report model-dependent orbit diagnostics in an adopted Milky Way potential. The orbit integrations interpret the observed slow state; they do not define the catalogue. We compare the low- V_{grf} sample against matched Gaia DR3 control samples drawn from higher velocity bands, and we treat external spectroscopic matches as context for the subset with available chemistry rather than as a population assignment for the catalogue as a whole.

Because this paper layers several kinds of statement, we distinguish them explicitly at the outset. *Direct Gaia observables*—sky position, parallax, proper motion, radial velocity, and the derived V_{grf} —define the catalogue. *Threshold-probability outputs*—the per-star $P(V_{\text{grf}} < 25 \text{ km s}^{-1})$ and the resulting tiers—are Monte Carlo membership scores computed from those observables and their uncertainties. *Orbit-model outputs*—eccentricity, R_{peri} , R_{apo} , z_{max} , energy, angular momentum, and actions—are deterministic functions of an

adopted Galactic potential, and are model-dependent rather than measured. *Exploratory sensitivity products*—barred-potential central reaches, long-apocentre inner-reach and Sgr A* counts, resonance overlays, and chaos diagnostics—probe robustness and are labelled as such throughout. The conclusions rest most heavily on the first two layers. When both point-estimate and Monte Carlo orbit products are quoted, the sampling scheme is labelled explicitly. The orbit-model layer also has a time horizon: the direction of a star’s next excursion follows from its measured velocity, but an integration carried far enough forward or backward describes the adopted potential rather than the star. We set that horizon quantitatively in Section 5 and confine per-star trajectory statements to within it.

Section 2 describes the sample definition, uncertainty propagation, controls, and adopted orbit model. Section 3 presents the empirical catalogue properties and chemical cross-matches. Section 4 gives the model-dependent dynamical results. Section 5 discusses the interpretation and its limits, and Section 6 summarises the conclusions.

2. Sample Definition and Methods

2.1. Gaia DR3 Sample

The parent source is Gaia DR3 (Gaia Collaboration et al. 2023), restricted to stars with full six-dimensional phase-space information: sky position, parallax, proper motion, and radial velocity. Radial velocities in DR3 are available for approximately 33.8 million stars observed by the Radial Velocity Spectrometer (Katz et al. 2023). This 6D subset has a non-trivial selection function in sky position, magnitude, and colour (Castro-Ginard et al. 2023; Cantat-Gaudin et al. 2023); it is not a neutral slice of the Galaxy, and distant 6D samples are commonly giant-heavy, with incompleteness expected for cool faint dwarfs.

The sample was constructed from a local mirror of the Gaia DR3 source table using the columns needed for a six-dimensional solution: `source_id`, `ra`, `dec`, `parallax`, `pmra`, `pmdec`, `radial_velocity`, the corresponding uncertainties and astrometric correlations, broad-band photometry, `ruwe`, and `duplicated_source`. The equivalent Gaia Archive selection is `radial_velocity IS NOT NULL, parallax > 0, pmra IS NOT NULL, pmdec IS NOT NULL, and parallax_over_error > 5`; the literal ADQL query is shipped as `ADQL_selection.sql`. No RUWE or duplicated-source cut is applied to define the point-estimate catalogue; those flags define the Gold subset below. Heliocentric observables are transformed into Galactocentric coordinates and velocities, and Galactic rest-frame speeds are computed.

To avoid imposing the final catalogue boundary during candidate construction, we first scanned the full local Gaia DR3 source mirror and retained an intentionally broad legacy buffer with preliminary $V_{\text{grf}} < 200 \text{ km s}^{-1}$. This buffer contains 5,867,654 unique Gaia source IDs. After the final parallax zero-point, photogeometric-distance, and Solar-frame conventions are applied, 2,755 sources have corrected point-estimate $V_{\text{grf}} < 25 \text{ km s}^{-1}$ within the 20,829-source near-threshold candidate pool. Monte Carlo propagation of the observables, followed by the population-prior correction of Section 3.4, then defines the catalogue tiers used throughout the paper: Tier A ($P > 0.95$, 173 stars), Tier A+B ($P > 0.84$, 276 stars), and Tier A+B+C ($P > 0.50$, 621 stars). The remaining 2,134 corrected point-estimate sources below 25 km s^{-1} have $P \leq 0.50$ and are retained as Tier D rather than as part of the primary catalogue. For

continuity with the deposited catalogue we also release the uncorrected forward scores, which select 289, 541, and 1,952 stars at the same three thresholds; the population-prior tiers are strict nested subsets of these.

This is not a volume-complete sample and it is not intended as a complete inventory of all such stars in the Milky Way. It is a Gaia DR3 6D catalogue defined by a simple instantaneous kinematic state and reported with explicit threshold-membership probabilities.

Table 1. Sample accounting under the final distance and Solar-frame conventions. The candidate pool is the Gaia DR3 6D source list recovered from the full-parent buffered scan; the corrected point-estimate row applies the Lindegren+2021 parallax zero-point, Bailer-Jones+2021 photogeometric distances, and the default Galactocentric frame before evaluating the $V_{\text{grf}} < 25 \text{ km s}^{-1}$ boundary. Tiers are mutually exclusive except for the explicitly cumulative Tier A+B and Tier A+B+C rows.

Set	N
Gaia DR3 6D candidate pool	20,829
Corrected point-estimate $V_{\text{grf}} < 25 \text{ km s}^{-1}$	2,755
Tier A, $P(V_{\text{grf}} < 25) > 0.95$	173
Tier B increment, $0.84 < P(V_{\text{grf}} < 25) \leq 0.95$	103
Tier A+B, $P(V_{\text{grf}} < 25) > 0.84$	276
Tier C increment, $0.50 < P(V_{\text{grf}} < 25) \leq 0.84$	345
Tier A+B+C, $P(V_{\text{grf}} < 25) > 0.50$	621
Tier D, point-estimate $V_{\text{grf}} < 25 \text{ km s}^{-1}$ and $P \leq 0.50$	2,134
Candidate-pool sources outside the corrected cut	18,074

To keep the role of each count unambiguous, Table 2 maps each tier and derived sample to the results it anchors. The **primary catalogue** for the paper is Tier A+B ($N = 276$); orbit-derived medians and population distributions are computed on the broader Tier A+B+C sample ($N = 621$). Selection-function and chemistry diagnostics are reported only as population-scale context, never as a redefinition of the primary catalogue. The leading rows of the released machine-readable primary catalogue are shown in Table 3.

2.2. Quality Control

Because the slowest stars are the most vulnerable to measurement error, we report conservative high-quality subsets in addition to the tiered catalogue, following the Gaia DR3 catalogue-validation guidance (Babusiaux et al. 2023). The Gold observational-quality subset is defined inside Tier A+B by $\text{RUWE} < 1.4$, an acceptable DR3 RVS quality flag, parallax signal-to-noise > 10 , and fractional distance uncertainty $\sigma_d/d < 0.15$. Dynamical robustness is reported separately in Section 4.5, where additional orbit-derived criteria are applied.

Operationally, “acceptable DR3 RVS quality” means the released `rv_quality` field is ok: the source is not flagged as poor by either $p(\chi^2_{\text{RV}}) < 0.01$ or `rv_expected_sig_to_noise` < 2 , and is not flagged as marginal by the large-amplitude/few-transit or faint out-of-template-temperature screens defined in the public pipeline script.

Several later diagnostics necessarily use narrower subsets. The α -classified spectroscopic subset is defined only by the availability of APOGEE or GALAH [Fe/H] and alpha-element abundances within Tier A+B+C; it is used for within-subset literature-region comparisons, not for whole-catalogue population fractions. The selection-function inverse-weighted sums are likewise reported only as observability context, because

many slow stars occupy sparse colour–magnitude cells where the Gaia DR3 RVS selection-function tool returns its prior mean. Table 4 reports how the primary counts respond to these conservative RVS-quality cuts.

Table 4. Sensitivity of the main orbit summaries to removing stars flagged as poor by the internal Gaia DR3 RVS-quality screen. Values, including median eccentricity, are point-estimate QA diagnostics; the qualitative compact, radial-orbit result is unchanged.

Sample	N	$\bar{R}_{\text{peri}}$ (pc)	$\bar{e}$
Tier A+B+C, all RV qualities	621	132	0.970
Tier A+B+C, excluding poor RVS	578	132	0.969
Tier A+B, all RV qualities	276	114	0.973
Tier A+B, excluding poor RVS	256	115	0.972

2.3. Velocity-Threshold Uncertainty

The catalogue is defined by threshold-membership probability rather than by a hard point estimate alone. For every source in the propagated candidate pool we draw Monte Carlo realisations of $(\varpi, \mu_{\alpha*}, \mu_{\delta})$ from the Gaia parallax–proper-motion covariance submatrix, holding sky position fixed because the positional uncertainties are negligible for the velocity scale considered here. We draw radial velocity independently from its quoted uncertainty, apply the Lindegren et al. (2021) zero-point correction, and draw distance from the asymmetric Bailer-Jones et al. (2021) photogeometric posterior, following the parallax-handling guidance of Luri et al. (2018). We then recompute V_{grf} for each realisation using the same Solar parameters and coordinate convention as the point-estimate catalogue. This explicit propagation is complementary to recent Gaia DR3 Cartesian catalogue work that compares second-order and Monte Carlo covariance propagation for the full RVS sample (Zhang et al. 2025).

Tier-fraction percentages reported below carry Beta($k + \frac{1}{2}, n - k + \frac{1}{2}$) Jeffreys 68% intervals (Cameron 2011); integer-count diagnostics (e.g., long-apocentre inner-reach counts) carry Gehrels (1986) Poisson 1σ bands.

The Monte Carlo scheme uses 500 baseline realisations per source, refines stars in the transition band $0.30 < P < 0.70$ to 5,000 realisations, and assigns 10,000 realisations to stars whose point-estimate V_{grf} lies within 2 km s^{-1} of the threshold. A separate 100-star convergence check compares 500 and 5,000 realisations. In this release, 1,623 stars receive the 5,000-realisation refinement and 1,261 receive the 10,000-realisation near-threshold refinement. We use the resulting $P(V_{\text{grf}} < 25 \text{ km s}^{-1})$ values to define Tier A ($P > 0.95$), Tier B increment ($0.84 < P \leq 0.95$), Tier C increment ($0.50 < P \leq 0.84$), and Tier D (corrected point estimate below 25 km s^{-1} but $P \leq 0.50$).

Because every star carries a membership probability, the score-implied number of contaminants in any tier follows directly as $\sum(1 - P)$ over its members. Which probability is used matters a great deal, and we therefore state both. Under the *forward* Monte Carlo scores the tiers would appear very clean: 94% purity for a 541-star Tier A+B and 73% for a 1,952-star Tier A+B+C. Those figures are biased high, because the forward score takes no account of how many stars are available at each true speed to scatter into the window; evaluated under the population prior of Section 3.4, the same two samples have purities of only 80% and 42%. We therefore define the tiers on the population-prior probability throughout. On that definition the purities are 99.0% for the 173-star Tier A core, 95.4% for the 276-star Tier A+B primary catalogue, and 78.6% for the 621-star Tier A+B+C extension, corresponding to 12.7, 12.7,

Table 2. Which sample anchors which result. The primary catalogue for the paper is Tier A+B ($N = 276$); orbit medians and population distributions use Tier A+B+C ($N = 621$). Every numerical claim in the paper names its anchoring sample explicitly.

Sample	Used for
Tier A ($N = 173$, $P > 0.95$)	Highest-confidence empirical subset; robustness anchor.
Tier A+B ($N = 276$, $P > 0.84$)	Primary catalogue ; Gold subsets; selection-function context.
Tier A+B+C ($N = 621$, $P > 0.50$)	Orbit-derived medians; broad catalogue characterisation.
α -classified ($N = 117$; not velocity-thresholded)	Splash/GSE/Aurora/disc fractions only.
Tier A+B SF-weighted	Population-scale context; never per-star weights.

Table 3. Sample rows from the Tier A+B primary catalogue. The full table is available in machine-readable form in the review package as `mrt/catalogue_tierAB_mrt.txt`; the FITS version is `catalogues/catalogue_expanded_tierAB.fits`.

Gaia DR3 source_id	R.A. (deg)	Decl. (deg)	V_{grf} (km s^{-1})	$P(V_{\text{grf}} < 25)$	N_{MC}	Tier	RVS QC
19320583963797632	39.10617	7.09534	19.11	0.968	500	B	ok
36729808698742528	57.56126	11.89217	20.89	0.908	500	B	ok
47180254402996480	61.88618	18.71729	14.48	0.994	500	B	ok
231283683952040960	58.35711	42.70816	14.45	0.966	500	B	ok
262676695131174784	77.21617	51.32057	8.86	0.984	500	B	ok
347579131898942336	26.99229	41.91899	15.37	0.962	500	B	ok
366349822607618944	7.18968	35.97100	17.53	0.954	500	B	ok
391429820279708288	9.76207	49.52606	15.46	1.000	500	A	ok

Notes. Only eight rows are shown for form and content. The electronic table contains all 276 Tier A+B sources.

and 133 score-implied false positives respectively. Tier A+B remains the recommended high-confidence working sample and Tier A+B+C the high-completeness extension.

The retiering does not reorder the catalogue. No star enters any tier that was not already in the corresponding forward-defined tier: the population-prior tiers are strict nested subsets, with rank correlations between the two scores of $\rho_S = 0.997$ and $\tau_K = 0.963$. The correction moves the thresholds rather than the ranking, and the released products carry both scores so that either definition can be reproduced.

The Tier A+B boundary is used for the primary catalogue because $P > 0.84$ corresponds approximately to a one-sided one-sigma probability threshold: these stars are more likely than not to remain below 25 km s^{-1} even after a one-sigma upward fluctuation in the propagated threshold variable. The three operating thresholds correspond to standard one-sided Gaussian-tail levels: $P > 0.95$ (Tier A) is approximately a one-sided 1.6σ threshold, $P > 0.84$ (Tier A+B) is approximately a one-sided 1σ threshold, and $P > 0.50$ (Tier A+B+C) is the more-likely-than-not boundary. Tier C is reported as the broader, lower-confidence catalogue extension rather than as part of the conservative primary sample. These Gaussian-tail correspondences are interpretive labels for the operating thresholds, not detection significances or empirical truth calibrations. We therefore describe the summed $1-P$ values as nominal internal score-implied contamination, and test that score with both a Monte Carlo self-consistency check and a truth-labelled mock diagnostic below. Central 68% intervals on the primary numerical claims are collected in Table 22.

Because the Bailer-Jones et al. (2021) photogeometric posterior is itself conditioned on Gaia astrometry, drawing the distance independently of the astrometric covariance is an approximation. Recent kinegeometric distance work combines parallaxes and proper motions with an explicit velocity prior (Bailer-Jones 2023); we do not adopt that posterior for this velocity-selected catalogue because the velocity prior would be entangled with the low- V_{grf} quantity being measured. The catalogue therefore uses the Bailer-Jones et al. (2021) posterior as an external asymmetric distance prior only to define

threshold-membership probabilities, and treats distance-method and covariance-coupling tests as sensitivity diagnostics rather than as a full joint posterior or as part of the catalogue definition.

We quantified this approximation with a stress test over the entire tier-sensitive band: all 3,342 candidates with adopted $P(V_{\text{grf}} < 25 \text{ km s}^{-1}) \geq 0.30$. The test keeps the BJ split-normal distance marginal but imposes a maximal Gaussian-copula anti-correlation between the parallax residual and the distance quantile, recomputing the independent- and coupled-distance probabilities at matched realisations so that ΔP isolates the coupling. The median $|\Delta P|$ is 0.009, the 84th percentile 0.021, the 95th 0.028, and the largest shift 0.055. Across the band, 114 of 3,342 sources change tier label; the full tier-label confusion matrix is given in Table E.1. The primary catalogue is robust to this worst-case coupling: all 276 Tier A+B members remain above $P = 0.84$ under both the independent and the copula-coupled treatment, with a median $|\Delta P|$ of 0.001 and a maximum of 0.023 across that sample—the population-prior thresholds sit far enough from the boundary that distance-prior coupling cannot move them. The largest shifts occur for near-threshold sources, which the probabilistic tiering already flags (Fig. 1).

As a column-level audit of distance-prior sensitivity, we also report the Gaia parallax signal-to-noise distribution for the tiered catalogue. In the primary Tier A+B sample, 19 of 276 stars (6.9%) lie in the lower-SNR $5 < \varpi/\sigma_\varpi \leq 10$ interval and 257 (93.1%) have $\varpi/\sigma_\varpi > 10$; none have a BJ 68% fractional distance half-width above 0.15. In the broader Tier A+B+C extension, where the lower-confidence Tier C dominates, 69 of 621 stars (11.1%) lie in the 5–10 interval, 552 (88.9%) exceed 10, and none have BJ fractional distance half-width above 0.15. Thus the potentially prior-sensitive distance regime is concentrated in the completeness extension, not in the primary catalogue; the copula test above is our stress bound on how this approximation affects threshold membership.

A separate Monte Carlo self-consistency reconstruction repeats the 500-realisation baseline pass with an independent random seed, selects the 2,275 sources with baseline $0.30 < P < 0.70$, and compares deciles of the baseline probability against

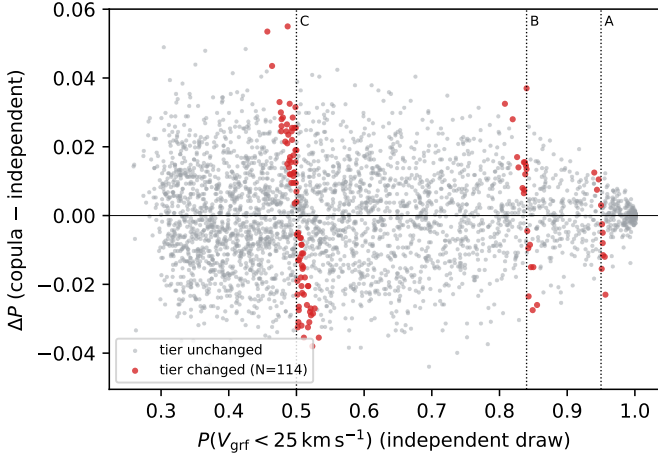


Fig. 1. Change in $P(V_{\text{grf}} < 25 \text{ km s}^{-1})$ when the BJ-style distance draw is coupled to the Gaia parallax residual, for the full tier-sensitive band (3,342 sources with adopted $P \geq 0.30$), plotted against the independent-draw probability. The stress test imposes a maximal copula anti-correlation and probes the distance-covariance approximation in the catalogue Monte Carlo; sources that change tier label are highlighted and the tier boundaries are marked. The full tier-label confusion matrix is given in Table E.1.

a held-out 5,000-realisation refinement. The decile medians lie close to the one-to-one relation: the median absolute decile residual is 0.0019 and the largest is 0.0040 (Fig. 2). This diagram tests numerical stability of the Monte Carlo score; it does not by itself establish empirical truth-calibration of the catalogue probabilities. It is distinct from the Gaia RVS selection-function systematic quantified separately in Section 5.

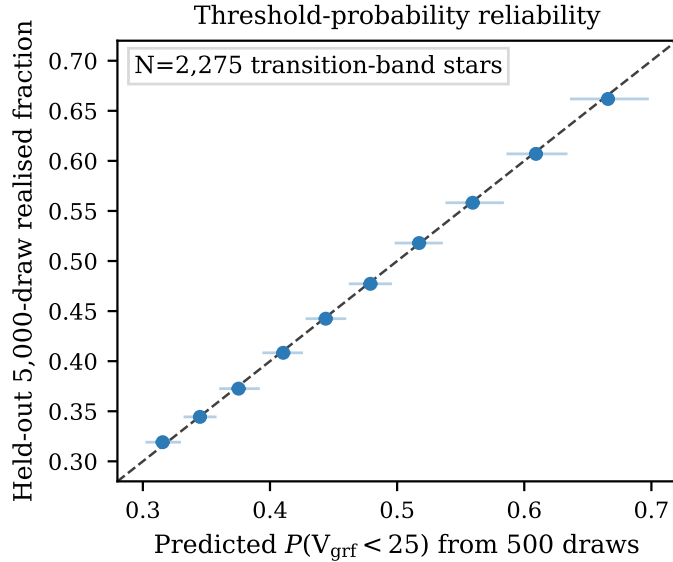


Fig. 2. Monte Carlo self-consistency of the threshold-membership probabilities in the transition band. Points bin stars by the recomputed 500-realisation baseline probability; the ordinate is the realised fraction in an independent held-out 5,000-realisation refinement. Error bars are central 68% Beta($k + \frac{1}{2}, n - k + \frac{1}{2}$) intervals (Cameron 2011).

The truth-labelled GeDR3mock diagnostic provides a complementary score check. Among mock stars that enter the same RVS and parallax-SNR pipeline, binning by predicted $P(V_{\text{grf}} < 25 \text{ km s}^{-1})$ gives a monotonic relation between the

score and the fraction of injected stars whose true speed is below 25 km s^{-1} (Table 5, Fig. 3). The pipeline-input Brier score is 0.031; the mock Tier A, Tier A+B, and Tier A+B+C score cuts have true sub-25 fractions of 0.997, 0.987, and 0.951, respectively. Because the mock is deliberately balanced across injected velocity bins, this is a truth-labelled score diagnostic rather than a population-prior calibration for the observed Gaia catalogue.

Table 5. Truth-labelled GeDR3mock probability-score diagnostic for pipeline-input mock stars. The mock is balanced across injected true V_{grf} bins, so this table checks monotonic score behaviour rather than population-prior calibration. The Brier score over the pipeline-input mock is 0.031.

$P(V_{\text{grf}} < 25)$ bin	N	$\langle P \rangle$	true sub-25 frac.	true- $\langle P \rangle$
0–0.10	19868	0.002	0.006	+0.004
0.10–0.30	897	0.192	0.368	+0.176
0.30–0.50	733	0.399	0.628	+0.229
0.50–0.70	843	0.602	0.805	+0.203
0.70–0.84	800	0.777	0.890	+0.113
0.84–0.95	1200	0.901	0.956	+0.055
0.95–1.00	3511	0.990	0.997	+0.007

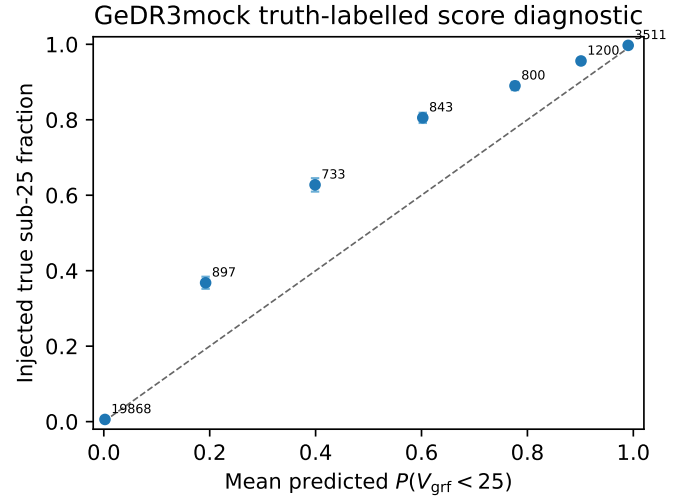


Fig. 3. Truth-labelled GeDR3mock probability-score diagnostic for pipeline-input mock stars. Points bin injected stars by their predicted $P(V_{\text{grf}} < 25 \text{ km s}^{-1})$ and compare with the fraction whose injected true V_{grf} is below 25 km s^{-1} . Numbers annotate the stars per bin. The mock prior is balanced across injected velocity bins, so the figure checks score monotonicity and high-probability behaviour rather than the population-prior calibration of the observed catalogue.

2.4. Galactocentric Transformation and Supporting Orbit Model

For the Galactocentric transformation we adopt $R_{\odot} = 8.178 \text{ kpc}$ (GRAVITY Collaboration et al. 2019), a Solar height above the plane of 25 pc, Solar peculiar motion $(U, V, W) = (11.1, 12.24, 7.25) \text{ km s}^{-1}$ from Schönrich et al. (2010), and $V_{\text{circ}} = 229.0 \text{ km s}^{-1}$ at the Solar radius, consistent with the Eilers et al. (2019) rotation-curve value ($v_c(R_0) = 229.0 \pm 0.2 \text{ km s}^{-1}$) and with the axisymmetrised Hunter et al. (2024) potential used for the orbit integrations. The coordinate transformation is performed with `astropy.coordinates` (Astropy Collaboration et al. 2013, 2018, 2022) and the

resulting Cartesian Galactocentric phase-space vectors are passed to *agama*. Three additional Solar-parameter variants from the recent literature (GRAVITY Collaboration 2022; Schönrich 2012; Reid & Brunthaler 2020) are used as a sensitivity bracket in Appendix A (Table A.5).

We apply the per-star Lindegren et al. (2021) Gaia parallax zero-point correction and use the Bailer-Jones et al. (2021) photogeometric distance posterior as the default distance estimator. The inverse-parallax distance, with and without the zero-point correction, is retained only as a sensitivity diagnostic. In the propagated candidate pool, the adopted photogeometric-distance convention gives 2,755 point-estimate sources below the cut before probability-tiering.

The default supporting orbit model is the static, axisymmetrized form of the analytic Milky Way potential of Hunter et al. (2024). The underlying Hunter+2024 model is not Gaia-DR3-calibrated; it combines analytic bulge/bar, nuclear, disc, and halo components constrained by rotation-curve and gas-terminal-velocity data. We use its axisymmetrized form as the catalogue default because it keeps orbital quantities such as E , L_z , and Stäckel-fudge actions well-defined. Orbit integration and action estimates use *agama* (Vasiliev 2019); actions are computed with the Stäckel-fudge formalism (Binney 2012; Sanders & Binney 2016). The orbit integrations are fully three-dimensional, evolving (x, y, z, v_x, v_y, v_z) . Throughout, $R(t) = \sqrt{x(t)^2 + y(t)^2}$ and the reported $R_{\text{peri}} = \min_t R(t)$ and $R_{\text{apo}} = \max_t R(t)$ are cylindrical radial turning points. Thus R_{peri} measures closest approach to the Galactic spin axis, not necessarily to Sgr A* as a point. Proximity to the Galactic centre is assessed separately using the spherical closest approach $r_{\text{sph,min}} = \min_t \sqrt{x(t)^2 + y(t)^2 + z(t)^2}$. Point-estimate integrations span 4 Gyr, and cylindrical pericentres, apocentres, and maximum heights are read from a 40,001-point, parabola-interpolated trajectory sampling; the 5,000-realisation orbit Monte Carlo uses the same 4 Gyr interval with 1,001 stored samples per realisation. This interval samples many radial periods for the compact inner-reaching orbits ($T_r \lesssim 100$ Myr at $R_{\text{peri}} \sim 100$ pc) while keeping the orbit products descriptive rather than evolutionary. A static-potential re-integration of a random 100-star Tier A subset to 8 Gyr changes the sample medians by only +5.5 pc in R_{peri} , +0.017 kpc in R_{apo} , and +0.039 kpc in z_{max} relative to the 4 Gyr products. This longer-time experiment is a convergence check only; the released orbit statistics use the full Tier A+B+C integrations. A rigidly rotating barred Hunter/Sormani variant (Sormani et al. 2022) is used as a sensitivity test for non-axisymmetric inner-Galaxy effects, not as the default catalogue definition. The static integrations conserve energy to a median relative drift of 6×10^{-7} ; in the rotating barred frame the inertial-frame energy is not conserved (median relative drift 2%, 99th percentile 12%), but the Jacobi integral $E_J = E - \Omega_p L_z$ is conserved to a median 5×10^{-7} , confirming integration fidelity.

In this manuscript, orbit-derived quantities such as eccentricity, R_{peri} , R_{apo} , z_{max} , energy, angular momentum, and actions are presented as outputs of the adopted model. They are informative, but they remain model-dependent, especially for stars whose inferred cylindrical pericentres lie within a few tens of parsecs of the Galactic spin axis. Pericentres, apocentres, and maximum heights are measured directly from the integrated trajectories, not inferred from Stäckel-fudge actions. The reported single-evaluation actions are accompanied by a full-sample convergent audit: for every Tier A+B+C star they are compared with orbit-time-averaged actions over the

4 Gyr integration. The azimuthal action $J_\phi \equiv L_z$ is recovered exactly and J_R to a median 8%, but Stäckel-fudge accuracy degrades for the vertical action J_z of these plunging orbits (Wright & Binney 2026): its median fractional difference is 62% over Tier A+B+C (85% in Tier A), and 463 of the 621 stars are flagged `sampled_poor`. The catalogue therefore ships orbit-time-averaged actions and a per-star action-reliability flag alongside the single-evaluation values, and we treat J_z -dependent statements as qualitative. This caveat is sharpened in Section 4.5 alongside the barred-potential sensitivity test.

2.5. Matched-Control Sample

To test whether the slow sample's orbit-derived properties are distinctive rather than generic to Gaia DR3 6D stars at matched observability, we constructed a multi-band velocity-stratified control sample from the full Gaia DR3 6D catalogue. Four velocity bands were defined: 25–50, 50–100, 100–200, and 200–260 km s^{-1} . Within each band, controls are kernel-weighted on the slow catalogue's broad observability marginals in G , $\log d$, and $\sin |b|$, with the same valid-radial-velocity requirement as the slow-star sample. The effective control samples contain 1,955, 2,586, 2,788, and 2,749 stars for the four bands respectively.

The adequacy of this weighting is audited rather than assumed. For nine observability covariates— G , $\log d$, $|\sin b|$, $\sin l$, $\cos l$, BP–RP, $\ln \text{RUWE}$, $\log_{10}(\varpi/\sigma_\varpi)$, and G_{RVS} —we compute the standardized mean difference (SMD) between each control band and the slow catalogue before and after weighting (Table 6; Fig. 4). The product-of-one-dimensional observability kernel does not by itself achieve covariate balance: residual |SMD| values remain at the 0.6–0.8 level and the matched marginals are not driven toward zero, because matching each marginal independently ignores the joint covariate correlations. We therefore adopt entropy balancing (Hainmueller 2012) as the principled weighted-inference layer; it solves for the maximum-entropy control weights that match every covariate mean to $|\text{SMD}| \lesssim 10^{-3}$, with band effective sample sizes of 387, 990, 1,438, and 1,474. The across-band ordering of median pericentre with V_{grf} (§4.2) is stable under all four weighting schemes we tested—unweighted, the observability kernel, propensity-score inverse-probability weighting, and entropy balancing—so the matched comparison is not an artefact of the weighting choice. These reweightings use the existing local control products rather than a fresh Gaia archive scan or a replacement control-parent construction.

The 25–50 km s^{-1} band yields fewer matched controls because the candidate pool at these velocities is intrinsically small: within the Gaia DR3 6D parent sample after the basic quality requirements used for control construction (valid radial velocity and parallax signal-to-noise > 5), there are only 4,763 stars in this range, compared with 39,853 at 50–100 km s^{-1} and over 1.4 million at 100–200 km s^{-1} . Overall, the low-velocity tail is extremely sparse within the observed Gaia DR3 6D catalogue: 7,622 stars below 50 km s^{-1} out of roughly 6.8 million stars in that quality-filtered 6D parent sample, or 0.11%. All controls were orbit-enriched using the same static Hunter+2024 potential and integration settings as the slow-star sample.

The stratified design tests how orbital properties vary with V_{grf} , rather than comparing the slow-star sample to a single control window. The 25–50 km s^{-1} band is the adjacent kinematic comparison to the slow-star sample; the 100–200 km s^{-1} band samples a mixture of thick-disc and inner-halo kinematics; and the 200–260 km s^{-1} band brackets the circular

Table 6. Covariate balance of the velocity-stratified matched controls relative to the Tier A+B+C slow catalogue, expressed as standardized mean differences (SMD). For each band the columns give the SMD before weighting (unweighted) and after entropy balancing; $|\text{SMD}| < 0.1$ indicates good balance. The paper’s product-of-1D observability kernel does not reduce these SMDs (Fig. 4), whereas entropy balancing matches every covariate mean to $|\text{SMD}| \lesssim 10^{-3}$. The footer gives the entropy-balanced effective sample size N_{eff} and weighted median cylindrical pericentre, which preserves its monotonic increase with V_{grf} .

covariate	25–50		50–100		100–200		200–260	
	before	ebal.	before	ebal.	before	ebal.	before	ebal.
G	+0.47	0.000	+0.35	0.000	+0.27	0.000	+0.26	0.000
$\log_{10}(d/\text{pc})$	+0.61	0.000	+0.43	0.000	+0.34	0.000	+0.30	0.000
$ \sin b $	-0.54	0.000	-0.35	0.000	-0.31	0.000	-0.35	0.000
$\sin l$	+0.03	0.000	+0.03	0.000	+0.02	0.000	+0.02	0.000
$\cos l$	+0.67	0.000	+0.44	0.000	+0.36	0.000	+0.37	0.000
$BP - RP$	+0.44	0.000	+0.33	0.000	+0.33	0.000	+0.32	0.000
$\ln \text{RUWE}$	-0.16	0.000	-0.12	0.000	-0.06	0.000	-0.08	0.000
$\log_{10}(\varpi/\sigma_{\varpi})$	-0.54	0.000	-0.33	0.000	-0.25	0.000	-0.22	0.000
G_{RVS}	+0.30	0.000	+0.23	0.000	+0.15	0.000	+0.15	0.000
$N_{\text{eff}} (\text{ebal.})$	387		990		1438		1474	
med. R_{peri} (pc)	206		592		2676		3974	

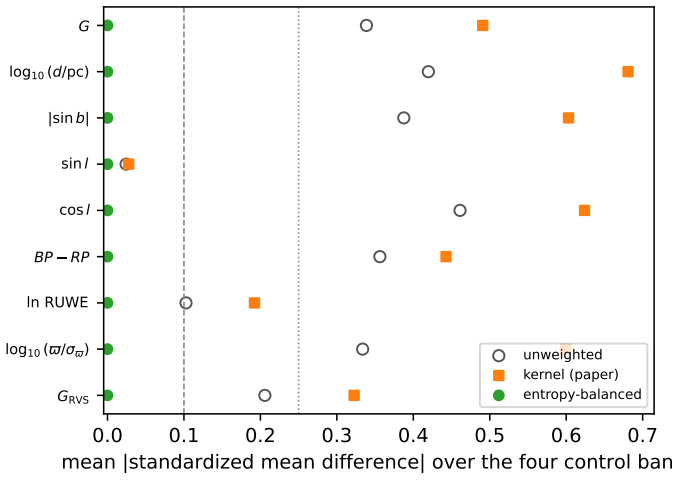


Fig. 4. Covariate balance (Love plot) of the velocity-stratified matched controls relative to the Tier A+B+C slow catalogue, averaged over the four control bands. Open circles are the unweighted standardized mean differences; orange squares are the paper’s product-of-one-dimensional observability kernel, which does not improve balance; green circles are entropy balancing, which matches every covariate mean to $|\text{SMD}| \approx 0$. The dashed and dotted lines mark the conventional 0.1 and 0.25 balance thresholds.

speed at $R_{\odot}$ (229 km s^{-1}) and therefore includes a substantial disc population.

2.6. Catalogue Contents and Availability

The accompanying catalogue includes Gaia `source_id`, the observables used for sample construction, adopted Galactocentric positions and velocities, V_{grf} , quality flags, and the principal orbit-derived quantities reported here (eccentricity, R_{peri} , R_{apo} , z_{max} , and related summary metrics). Orbit-derived fields are explicitly labelled as model-dependent outputs under the adopted potential rather than as direct Gaia observables. Per-star chemistry is included where available—Gaia GSP-Phot [M/H] and, for the APOGEE DR17/GALAH DR3 cross-match, spectroscopic [Fe/H], an α -abundance proxy, and a chemodynamic-population label, kept separate from the photometric metallicity—and the orbit catalogue adds both

single-evaluation and orbit-time-averaged actions with a per-star action-reliability flag.

3. Empirical Results

3.1. The Sample

The final catalogue is tiered. The Gaia DR3 6D candidate pool contains 20,829 near-threshold sources retained from a broad full-parent scan; under the final distance and Solar-frame conventions, 2,755 have corrected point-estimate $V_{\text{grf}} < 25 \text{ km s}^{-1}$. Point-estimate membership scales smoothly with the cut—1,420 and 4,594 sources fall below 20 and 30 km s^{-1} respectively—so the 25 km s^{-1} boundary (about 10% of the 229 km s^{-1} circular speed) is a convenience cut on a continuous slow tail, which is why membership is reported as probabilistic tiers rather than a hard threshold. The primary Tier A+B catalogue contains 276 stars with $P(V_{\text{grf}} < 25 \text{ km s}^{-1}) > 0.84$, while the broader Tier A+B+C catalogue contains 621 stars with $P > 0.50$. Unless otherwise stated, descriptive orbit-derived medians below are reported for Tier A+B+C and high-confidence observational-quality follow-up quantities for Tier A+B.

As a frame-dependent present-day phase-space check, we classified each star by its spherical Galactocentric radial velocity, $v_{r,\text{GC}} = (\mathbf{r}_{\text{GC}} \cdot \mathbf{v}_{\text{GC}})/|\mathbf{r}_{\text{GC}}|$. Negative values are inward, toward the Galactic centre, and positive values are outward. Table 7 reports this count both from the point-estimate phase-space solution and from a 5,000-realisation present-day Gaia Monte Carlo. This diagnostic uses the adopted Galactocentric frame but no orbit integration or Galactic potential. The check is included only to show that the sample does not have a large inward/outward radial-phase imbalance; the Monte Carlo probabilities are conditional on the measured catalogue and are not selection-function corrected significance tests.

Table 7. Frame-dependent present-day Galactocentric radial-phase check for the low- V_{grf} catalogue.

Sample	N	N_{in}	N_0	N_{out}	MC f_{in}	$P_{\text{in}>\text{out}}$
Tier A+B	276	124	40	112	$52.9^{+1.2}_{-1.2}\%$	0.989
Gold A+B	226	107	32	87	$55.7^{+1.3}_{-1.3}\%$	> 0.999
Tier A+B+C	621	281	80	260	$51.8^{+0.9}_{-0.8}\%$	0.977

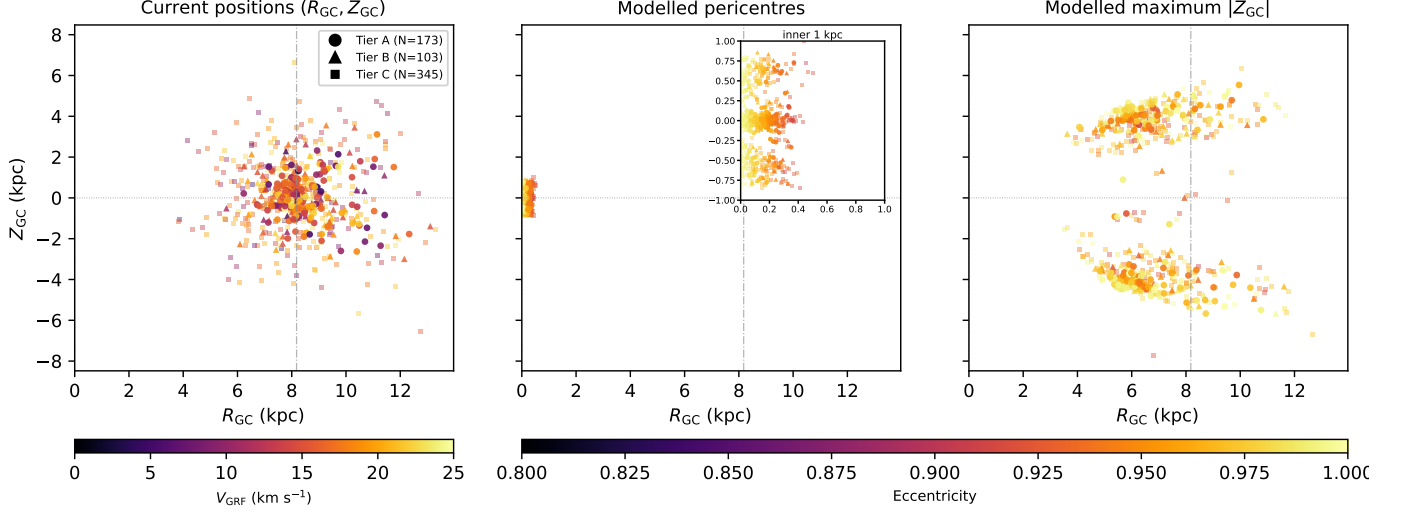


Fig. 5. Edge-on orbital context for the Tier A+B+C slow catalogue ($N = 621$) in R_{GC} vs. Z_{GC} . Current positions are observed coordinates, while the modelled cylindrical-pericentre and modelled maximum- $|Z_{GC}|$ panels are model-derived locations in the adopted static Hunter+2024 potential. This broader Tier A+B+C sample is used for orbit-context visualisation; the Tier A+B primary catalogue is distinguished from the Tier C extension by marker shape. The panels show current positions, modelled cylindrical-pericentre positions, and modelled maximum- $|Z_{GC}|$ points. Current positions are colour-coded by V_{grf} ; modelled cylindrical-pericentre and maximum- $|Z_{GC}|$ positions are colour-coded by eccentricity, with the pericentre panel inset expanding the inner kiloparsec. The right panel shows the vertical envelope of the adopted model trajectories; apocentre radii are reported separately in Table 18 and Fig. 13. Marker shape encodes tier: circles (Tier A), triangles (Tier B), squares (Tier C). The vertical dot-dashed line marks $R_{\odot} = 8.178$ kpc and the Galactic centre is at $R_{GC} = 0$. Because R_{GC} is cylindrical radius, these panels show distance from the Galactic spin axis rather than signed X_{GC} position. The current-position panel therefore folds signed side-of-centre information into positive radius; the pericentre panel and inset likewise show cylindrical spin-axis reach, not the signed side of the Galactic centre on which each passage occurs. The signed current-position and projected-pericentre geometry is retained in the face-on view of Fig. 6.

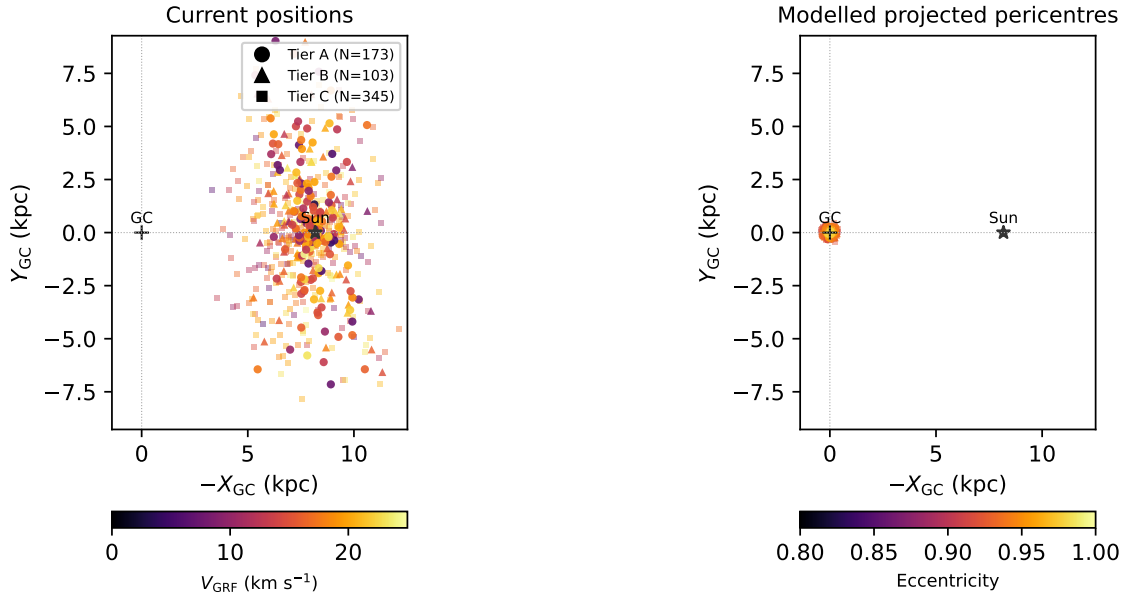


Fig. 6. Face-on view of the Tier A+B+C slow catalogue ($N = 621$) in the displayed $-X_{GC}$ vs. Y_{GC} plane, so that the Galactic centre is at left and the Sun lies to the right. As in Fig. 5, the broader Tier A+B+C orbit-summary sample is shown, with marker shape separating the Tier A+B primary catalogue from the Tier C extension. *Left:* current Galactocentric positions, colour-coded by V_{grf} ; the catalogue is observed in a Sun-centred local volume. *Right:* modelled, projected cylindrical-pericentre passage positions in the static Hunter+2024 potential, colour-coded by orbital eccentricity and shown on the same spatial scale as the left panel. Marker shape encodes tier (circle/triangle/square = A/B/C). The shared scale makes the cylindrical inner-Galaxy reach of the catalogue directly visible: all point-estimate cylindrical pericentres fall inside $R_{GC} < 1$ kpc in this fine trajectory sampling, with a median Tier A+B+C cylindrical pericentre of 115.5 pc in the 40,001-point interpolated readout. Pericentre is a turning point only in Galactocentric radius, not a physical “bounce”: the velocity vector does not instantaneously reverse, and highly radial trajectories generally traverse the inner Galaxy towards apocentre on the far side.

Note. N_{in} , N_0 , and N_{out} are point-estimate counts for spherical Galactocentric $v_r < -2 \text{ km s}^{-1}$, $|v_r| \leq 2 \text{ km s}^{-1}$, and $v_r > 2 \text{ km s}^{-1}$, respectively. MC f_{in} is the median and 16th–84th percentile interval of $N_{\text{in}}/(N_{\text{in}} + N_{\text{out}})$ from 5,000 present-day Gaia phase-space realisations per star. This frame-dependent diagnostic uses no orbit integration or Galactic potential. $P_{\text{in}>\text{out}}$ is conditional on the measured catalogue and Gaia uncertainties; it is not a selection-function-corrected significance test.

3.2. The Primary Catalogue Survives Conservative Quality Cuts

The Gold observational-quality subset starts from the Tier A+B primary catalogue and then applies only observational quality cuts. The catalogue Gold subset contains 226 stars after requiring RUWE < 1.4 , acceptable RVS quality, parallax signal-to-noise > 10 , and $\sigma_d/d < 0.15$. This gives a compact follow-up list whose definition does not depend on the orbit model.

Because these cuts are applied to a sample that is itself selected at a kinematic extreme, it is worth establishing explicitly that they are not circular. Two properties make that case, following the argument set out for age-extremum samples by Banik et al. (2026). First, every cut is defined on an axis distinct from V_{grf} : astrometric goodness of fit, image-parameter multiplicity, RVS template quality, parallax signal-to-noise, and fractional distance precision are all properties of the observation rather than of the velocity used to select the sample. Second, and more directly testable, the pass rate of each cut is essentially flat in V_{grf} across the full 20,829-source candidate pool (Table 8). For RUWE, image-parameter multiplicity, RVS quality, and fractional distance error, the largest difference in pass rate between any two V_{grf} bins spanning 0–50 km s^{-1} is 0.009, 0.011, 0.019, and 0.031 respectively; a cut that is this nearly blind to V_{grf} cannot preferentially manufacture slow stars. The one partial exception is the parallax signal-to-noise requirement $\varpi/\sigma_\varpi > 10$, which shows a larger spread (0.053) and, when applied alone, removes the largest share of the catalogue, retaining 88.9% of Tier A+B+C. A poorly measured parallax is one route to a spuriously low speed, so a cut that is mildly V_{grf} -dependent in this direction is expected, and we report it rather than treat it as neutral. In every case the cuts only ever remove candidates, so no cut can inflate the catalogue; and the median V_{grf} of the survivors moves by at most 0.06 km s^{-1} under any single cut, and by -0.05 km s^{-1} when all are applied together.

Unresolved binaries and problematic single-epoch radial velocities can perturb Gaia RVS velocities on scales comparable to the 25 km s^{-1} threshold. We therefore treat RUWE and RVS-quality screens as catalogue diagnostics rather than proof of single-star centre-of-mass velocities. In the catalogue products, 14/276 Tier A+B stars ($5.1^{+1.5}_{-1.2}\%$) and 32/621 Tier A+B+C stars ($5.2^{+1.0}_{-0.8}\%$) have RUWE > 1.4 (Jeffreys 68% intervals throughout); the Gold subset removes these systems, and the poor-RVS exclusion test in Section 4.5 shows that the compact radial-orbit summary is unchanged after removing flagged RVS solutions.

A direct check against the Gaia DR3 non-single-star catalogue sharpens this screen. Cross-matching by source_id against gaiadr3.nss_two_body_orbit flags 5 of the 276 Tier A+B sources (1.8%) and 13 of the 621 Tier A+B+C sources (2.1%) as resolved binaries—predominantly single-lined spectroscopic (SB1) solutions in the full Tier A+B+C extension—which are marked in the released catalogue. RUWE and the non-single-star tables are most sensitive to unresolved and to fully resolved companions respectively, which leaves the

marginally resolved regime between them. We therefore also apply the Gaia image-parameter cut `ipd_frac_multi_peak` ≤ 1 , which limits the percentage of focal-plane transits in which a source is split into multiple peaks (Banik et al. 2026; Pace et al. 2022, adopt a threshold of 2). This retains 265/276 (96.0%) of Tier A+B and 595/621 (95.8%) of Tier A+B+C, and it is largely independent of the RUWE screen, which removes a comparable but different 5.1% of Tier A+B. Applying all observational cuts jointly, including this one, leaves 220 Tier A+B stars—six fewer than the 226-star Gold subset defined without it—and 472 Tier A+B+C stars, with the median V_{grf} moving by -0.08 and -0.05 km s^{-1} respectively, so the surviving sample is if anything marginally slower rather than faster. Beyond these catalogued binaries, the RV-specific quality indicators bound the residual contamination (Table 9): the median DR3 RV uncertainty is 2.4 km s^{-1} for Tier A+B, 58% of primary-catalogue sources exceed 2 km s^{-1} , and 6.1% have an RV-constancy χ^2 p -value below 0.01 indicative of RV variability. Because the median uncertainty is a non-negligible fraction of the 25 km s^{-1} threshold, the $\sigma_{\text{RV}} < 2 \text{ km s}^{-1}$ cut (Section 4.5) and the Gold subset are the operative defences against RV-driven false positives.

We also queried public APOGEE DR17 and GALAH DR3 radial velocities for the Tier A+B+C source IDs as a sparse external consistency audit (Table 10). The overlap is footprint-selected rather than complete: 67 Tier A+B+C stars have APOGEE RVs and 62 have GALAH RVs. The Gaia-minus-external median offsets are small (-0.10 km s^{-1} for APOGEE and $+0.16 \text{ km s}^{-1}$ for GALAH), with robust scatter of 1.10 and 1.72 km s^{-1} , respectively. A minority of overlaps show $|\Delta \text{RV}| > 5 \text{ km s}^{-1}$ (6.0% for APOGEE, 9.7% for GALAH), so the external check supports the bulk RV scale but does not replace follow-up confirmation for individual boundary candidates.

In this section we restrict the robustness check to empirical observables and simple descriptive properties of the sample. The corresponding effect of the Gold cut on the model-derived orbit summaries is reported separately in Section 4.5.

3.3. Velocity-Threshold Membership Is Probabilistic

The corrected point-estimate catalogue contains 2,755 stars below $V_{\text{grf}} = 25 \text{ km s}^{-1}$, but the threshold itself is not equally secure for every star. After the population-prior correction this gives 173 stars in Tier A ($P > 0.95$), 276 stars cumulatively in Tier A+B ($P > 0.84$), and 621 stars cumulatively in Tier A+B+C ($P > 0.50$). The 2,134 corrected point-estimate members with $P \leq 0.50$ form Tier D. The paper therefore uses Tier A+B as the primary catalogue and Tier A+B+C for broader population-level orbit summaries. Table 12 summarises the tier definitions and counts, and Fig. 7 shows the membership probability as a function of the corrected point-estimate V_{grf} across the propagated pool.

A natural question is whether the low- V_{grf} population is more numerous than a smooth continuation of the Gaia DR3 velocity field would imply. We do not attempt to answer it with a count, and it is worth being explicit about why. Any such ratio requires a parent normalisation, and the matched-control library cannot supply one: it is constructed with comparable numbers of stars in each velocity band, so its sampling rate relative to the Gaia 6D catalogue varies by more than two orders of magnitude between the slowest and fastest bands, and its pooled velocity distribution is a property of that construction rather than of the Galaxy. Normalising a predicted count to the control library

Table 8. Observational quality cuts are orthogonal to the V_{grf} selection axis. “Pass-rate spread” is the largest difference in pass rate between any two V_{grf} bins spanning 0–50 km s^{−1} across the full 20,829-source candidate pool; a small spread means the cut is effectively blind to V_{grf} and therefore cannot manufacture the slow sample. Survival columns give the fraction of each tier retained.

Cut	Pass-rate spread (0–50 km s ^{−1})	Survival	
		Tier A+B	Tier A+B+C
RUWE < 1.4	0.009	262/276 (94.9%)	589/621 (94.8%)
ipd_frac_multi_peak ≤ 1	0.011	265/276 (96.0%)	595/621 (95.8%)
RVS quality ok	0.019	256/276 (92.8%)	578/621 (93.1%)
$\varpi/\sigma_{\varpi} > 10$	0.053	257/276 (93.1%)	552/621 (88.9%)
$\sigma_d/d < 0.15$	0.031	276/276 (100.0%)	621/621 (100.0%)
All cuts jointly	—	220/276 (79.7%)	472/621 (76.0%)

Table 9. Radial-velocity quality and unresolved-multiplicity audit by tier. σ_{RV} is the DR3 RV uncertainty; $f(\sigma_{\text{RV}} > 2)$ the fraction above 2 km s^{−1}; f_{var} the fraction with $p(\chi^2_{\text{RV}}) < 0.01$ (an RV-variability / binary indicator); $f_{\text{RUWE} > 1.4}$ the astrometric-binary fraction; and N_{NSS} the number matched in the Gaia DR3 non-single-star catalogue (nss_two_body_orbit).

tier	σ_{RV}	$f(\sigma_{\text{RV}} > 2)$	$\tilde{N}_{\text{tr}}$	f_{var}	$f_{\text{RUWE} > 1.4}$	N_{NSS}
A	2.19	54%	20	5.9%	4.2%	7
B	2.74	63%	21	6.3%	6.3%	2
A+B	2.37	58%	20	6.1%	5.2%	9
A+B+C	2.54	60%	18	5.4%	3.1%	23

Table 10. External radial-velocity consistency audit for the subset of Tier A+B+C stars with APOGEE DR17 or GALAH DR3 line-of-sight velocities. ΔRV is Gaia DR3 RVS minus the external survey velocity; the > 5 column gives the fraction with $|\Delta RV| > 5$ km s^{−1}. The coverage is footprint-selected and is used only as a robustness check.

survey	N_{ABC}	N_{AB}	med. ΔRV	scatter	> 5
APOGEE	67	20	-0.10	1.10	6.0%
GALAH	62	26	0.16	1.72	9.7%

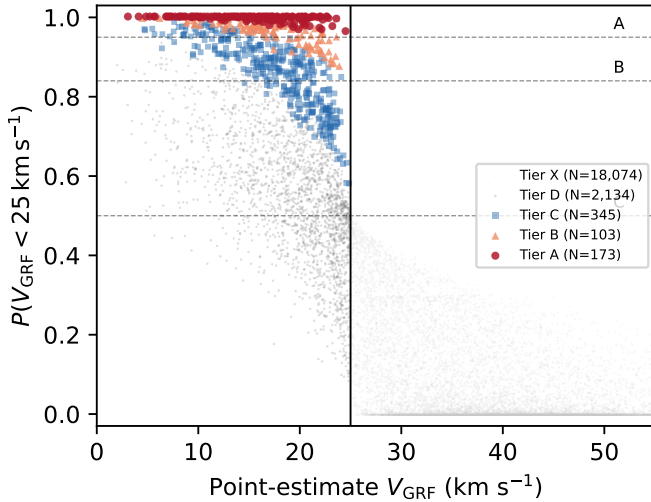


Fig. 7. Velocity-threshold membership probability versus corrected point-estimate V_{grf} for the propagated Gaia DR3 candidate pool. The horizontal dashed lines mark the Tier A, Tier A+B, and Tier A+B+C probability boundaries; the vertical line marks the 25 km s^{−1} point-estimate selection boundary. The catalogue is defined by threshold probability, not by a hard point estimate alone.

while counting observed slow stars from the complete parent scan compares two different populations. More fundamentally, Gaia RVS is magnitude limited, so this catalogue consists of giants and bright nearby dwarfs and contains essentially none of the M dwarfs that dominate the stellar census; no absolute density statement is available to us at any level of effort.

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The latent reconstruction of Section 3.4 answers the well-posed part of the question directly, and without a normalisation. Reconstructed from the complete near-threshold pool using each star’s own uncertainties, the latent V_{grf} distribution rises smoothly and monotonically through the selection boundary: the curvature of $\ln P$ stays below 0.02 per bin across 20–30 km s^{−1}, against a maximum of 0.08 elsewhere in the reconstructed range. The 25 km s^{−1} threshold leaves no imprint on the recovered distribution. The low- V_{grf} catalogue is therefore a tail selection from a smooth velocity distribution, not a physically distinct population separated by a discontinuity, and we make no claim of a number excess. The matched-control comparison is retained in Section 4.2 as a distributional check on orbital properties, where no parent normalisation is required.

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3.4. The Population Prior Halves the Expected Slow Count

The tier probabilities defined above are *forward* scores: for each star, the fraction of Monte Carlo realisations of its own measurement errors that fall below 25 km s^{−1}. They carry no information about the parent population. That omission matters here, because the candidate pool rises steeply with V_{grf} —188 sources below 10 km s^{−1} against 7,635 between 40 and 50 km s^{−1}—so far more stars are available to scatter down into the slow window than up out of it. This is the classical Eddington bias (Eddington 1913), and the forward scores do not correct for it. The injection-recovery calibration of Section 4.7 does not close the gap either, because its mock uses a balanced injected velocity prior by construction.

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We therefore reconstruct the latent V_{grf} distribution of the candidate pool directly. Writing L_{ij} for the Monte Carlo likelihood of star i in latent bin j , and P_j for the latent

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Table 11. “Gold observational-quality” subset: Tier A+B stars satisfying conservative observational quality cuts only — RUWE < 1.4, RVS quality flag = “ok” (Katz et al. 2023), parallax SNR > 10, $\sigma_d/d < 0.15$, and Tier A+B membership. Dynamical assumptions do not enter this subset; it is intended for observational follow-up that does not depend on the adopted potential.

Selection criterion	Count
Tier A+B ($P > 0.84$) base	276
& RUWE < 1.4	262
& RVS quality = “ok”	245
& parallax SNR > 10	226
& $\sigma_d/d < 0.15$	226
Final “gold observational-quality”	226
Median V_{grf} (km s ⁻¹)	16.0
Median G (mag)	13.4
Median distance (pc)	1,698

Table 12. Monte Carlo threshold-membership probabilities for the $V_{\text{grf}} < 25$ km s⁻¹ cut. Astrometric realisations use the Gaia DR3 parallax–proper-motion covariance, with sky position held fixed, plus an independent radial-velocity error, with distance drawn from the asymmetric Bailer-Jones+2021 photogeometric posterior. The candidate pool contains 20,829 Gaia DR3 sources; after the final correction and distance convention, 2,755 have corrected point-estimate $V_{\text{grf}} < 25$ km s⁻¹. “Sources newly admitted beyond the legacy preselection” is the count of Tier A+B+C members that do not appear in the pre-v15 slow-velocity preselection list maintained during catalogue development; they are the additions enabled by the zero-point-corrected, propagated, probability-tiered pipeline reported here.

Diagnostic	Value
Candidate pool propagated through MC	20,829
Corrected point-estimate $V_{\text{grf}} < 25$ km s ⁻¹	2,755
$P(V_{\text{grf}} < 25 \text{ km s}^{-1}) > 0.95$ (Tier A)	173
$0.84 < P(V_{\text{grf}} < 25 \text{ km s}^{-1}) \leq 0.95$ (Tier B increment)	103
$P(V_{\text{grf}} < 25 \text{ km s}^{-1}) > 0.84$ (Tier A+B; primary)	276
$0.50 < P(V_{\text{grf}} < 25 \text{ km s}^{-1}) \leq 0.84$ (Tier C increment)	345
$P(V_{\text{grf}} < 25 \text{ km s}^{-1}) > 0.50$ (Tier A+B+C)	621
Tier D (point-estimate < 25 km s ⁻¹ , $P \leq 0.50$)	2,134
Median $P(V_{\text{grf}} < 25 \text{ km s}^{-1})$ for Tier A+B+C	0.80
Sources newly admitted beyond the legacy preselection (Tier A+B+C)	200

distribution, we iterate

$$w_{ij} = \frac{P_j L_{ij}}{\sum_k P_k L_{ik}}, \quad P_j = \frac{1}{N} \sum_i w_{ij}, \quad (1)$$

to convergence. This is the iterative rectification of an observed distribution introduced by Richardson (1972) and Lucy (1974), equivalently expectation–maximisation (Dempster et al. 1977); in the Gaia era the same deconvolution problem is more often approached parametrically, as in the extreme-deconvolution formalism of Bovy et al. (2011). We adopt the non-parametric binned form because the quantity of interest is the shape of the distribution near a threshold rather than a mixture description of it. Our implementation follows the recent application of this approach by Banik et al. (2026) to the analogous problem of the oldest-star age extremum, which is where we encountered it in this context. The converged w_{ij} are per-star posteriors with the population prior folded in, so $\sum_{V_j < 25} w_{ij}$ is a revised membership probability for star i . The likelihood grids come from re-running the Section 2.3 forward model at 1 km s⁻¹ resolution, which reproduces the released per-star probabilities to within Monte Carlo noise (summed P of 2,556.4 against 2,555.9). The median per-star V_{grf} uncertainty is 8.5 km s⁻¹, with a 5–95 percentile range of 1.7–21.1 km s⁻¹.

Unregularised (1) is a maximum-likelihood deconvolution and develops the bin-to-bin oscillations characteristic of that algorithm at high iteration count, exactly as Banik et al. (2026) report for their unsmoothed variant. We therefore apply a Gaussian smoothing kernel to P_j after each iteration, the smoothed-EM regularisation of Silverman et al. (1990), and scan its width. The oscillations are confined to the 35–50 km s⁻¹ region; below 25 km s⁻¹ the reconstructions for kernel widths of

0, 0.75, 1.5, and 3 bins are indistinguishable, and the recovered count is stable from roughly the tenth iteration onwards (Fig. 8).

The result is a substantial downward revision, summarised in Table 13. The expected number of pool members with true $V_{\text{grf}} < 25$ km s⁻¹ falls from the naive sum of forward probabilities, 2,556, to 1,234⁺¹¹⁰₋₁₄₆, and the revised tier counts fall from 289, 541, and 1,952 to approximately 173, 276, and 621. Two features make this robust. The reconstruction is insensitive to the latent-grid ceiling, giving 1,186 to 1,198 expected slow stars for ceilings of 40 to 80 km s⁻¹; and the full spread across smoothing kernels, 1,198 to 1,365, is comparable to the bootstrap uncertainty.

Three qualifications belong with this number. First, the correction is conservative: the candidate pool was screened at a preliminary $V_{\text{grf}} < 50$ km s⁻¹ and is complete only below that, so latent bins above the screen are under-populated relative to the real Galaxy and down-scatter from genuinely fast stars is under-represented. The true contamination can only be larger than reported here. Second, we treat the Monte Carlo posterior of each star as its likelihood, which is exact under a flat prior in V_{grf} and approximate given the Bailer-Jones et al. (2021) distance prior; the same approximation is made by Banik et al. (2026). Third, and most importantly for the released products, the correction is close to monotonic in the forward score, so the ranking of the catalogue is preserved even though its absolute calibration shifts. No star enters a population-prior tier that was not already in the corresponding forward tier, and the two scores have rank correlations of $\rho_S = 0.997$ and $\tau_K = 0.963$. Retiering therefore moves the thresholds without reordering the catalogue, which is why we adopt the population-prior scores as the tier definition (Section 3.3) rather than carrying the

Table 13. Effect of folding the population prior into the threshold-membership probabilities. “Forward” counts use the released per-star Monte Carlo scores $P(V_{\text{grf}} < 25 \text{ km s}^{-1})$, which carry no information about the parent population. “Population-prior” counts come from the latent-distribution reconstruction of Sec. 3.4, which corrects for the fact that far more stars are available to scatter down into the slow window than up out of it. Nominal values use a smoothing kernel of 1.5 bins; uncertainties are 68% intervals from 200 bootstrap resamplings over stars. Varying the kernel over 0–3 bins moves the tier counts to 168–174, 267–277, and 619–630, and the expected count to 1,198–1,365; that spread is a systematic rather than a statistical one and is comparable to the bootstrap width. Forward tier counts are the released values; the reconstruction re-runs the Monte Carlo and reproduces them to within its sampling noise (289/548/1,951).

Sample	Forward	Population-prior
Tier A	289	173^{+15}_{-16}
Tier A+B	541	276^{+18}_{-32}
Tier A+B+C	1,952	621^{+38}_{-45}
Expected $N(V_{\text{grf}} < 25 \text{ km s}^{-1})$	2,556	$1,234^{+110}_{-146}$

correction only as a caveat: the forward thresholds of 0.95, 0.84, and 0.50 correspond to population-prior thresholds of 0.987, 0.957, and 0.813, so the released tiers were simply cut too loosely. Both scores ship with the catalogue, so either definition can be reproduced. Because the tiers are nested, every orbit product quoted below was already computed for every retained star; the retiering re-aggregates those products over a subset and integrates no new orbits.

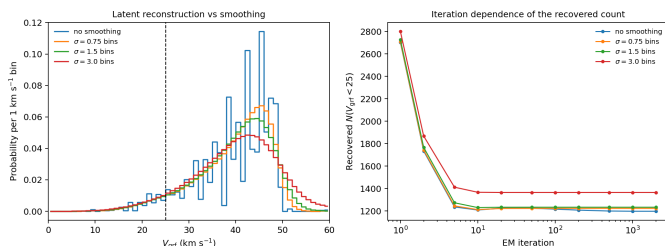


Fig. 8. Latent V_{grf} reconstruction from Eq. (1). *Left:* the recovered latent distribution for Gaussian smoothing kernels of 0, 0.75, 1.5, and 3 bins. The unsmoothed solution oscillates between adjacent bins over 35–50 km s^{-1} , the familiar noise amplification of maximum-likelihood deconvolution, but all four agree closely below the 25 km s^{-1} threshold (dashed line), which is the region that sets the reported counts. The sharp fall above 50 km s^{-1} is the candidate pool’s screening edge, not a feature of the Galaxy. *Right:* the recovered $N(V_{\text{grf}} < 25 \text{ km s}^{-1})$ against iteration number. The count converges by roughly the tenth iteration and is then flat for every smoothing choice, so the revision is not an artefact of the iteration cap.

3.5. Chemistry Does Not Define the Sample

Metallicity information is used here only as a diagnostic of chemical heterogeneity and estimator bias, not for population assignment. The metallicity spread in the matched subset is broad enough that the catalogue should not be treated as a single chemical population.

These official Gaia DR3 GSP-Phot metallicities are low-resolution photometric estimates with documented systematics (Andrae et al. 2023; Recio-Blanco et al. 2023): in particular, GSP-Phot tends to overestimate metallicity for cool, metal-poor giants by 0.2–0.5 dex, which biases the present sample toward

solar values. In the catalogue, GSP-Phot metallicities are finite for 1,279 Tier A+B+C stars and have a median $[M/H] = -0.24$ (Fig. 9). The spectroscopic α -subset discussed below is more metal-poor, with median $[\text{Fe}/H] = -0.81$. This offset is not interpreted as a population difference: it reflects both the known GSP-Phot metallicity systematics and the strongly non-random targeting of the APOGEE/GALAH overlap. The Zhang et al. (2023) XP catalogue remains the preferred larger photometric metallicity comparison for a future broader cross-match, but the present paper uses only the synchronized GSP-Phot and spectroscopic products as release materials.

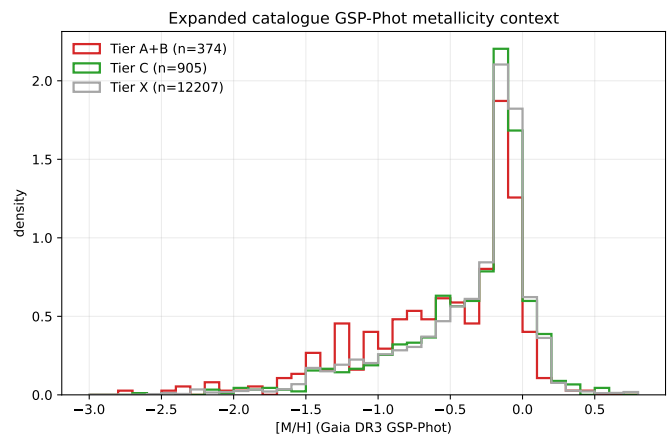


Fig. 9. Gaia GSP-Phot metallicity distribution for the catalogue, shown as a low-resolution contextual diagnostic. These values are not used for population-region assignment.

3.6. Chemical Properties from Spectroscopic Cross-Matching

We cross-matched the full 20,829-source candidate pool against APOGEE DR17 (Abdurro’uf et al. 2022) and GALAH DR3 (Buder et al. 2021) using exact Gaia source identifiers rather than positional matching. The query returns 604 APOGEE matches and 489 GALAH matches across the candidate pool. Spectroscopic abundances are rare here: 66 stars have usable APOGEE α -element abundances and 59 have usable GALAH abundances, and after survey-priority deduplication 117 stars carry the joint $[\text{Fe}/H] + \alpha$ combination required for the literature-region comparisons. Restricting that set to the stars above the adopted velocity threshold would leave only 48.

We do not impose that restriction. The velocity threshold is not a chemical criterion, and a star does not stop being chemically informative because its rest-frame speed is more likely 30 km s^{-1} than 22 km s^{-1} . We therefore report the chemistry on the full 117-star α -classified sample and quantify the membership difference per region rather than cutting on it (Table 15): alongside the raw count we give ΣP_{pop} , the summed population-prior probability, which is the effective number of genuinely slow stars each region contributes, and the count surviving the $P > 0.5$ threshold. The summed probability over the whole α -classified sample is 58.5.

The three treatments give consistent proportions—Splash 43.6%, 44.6%, 45.8%; Gaia-Sausage/Enceladus 19.7%, 17.9%, 18.8%; Aurora 18.8%, 19.5%, 20.8% by raw count, by summed probability, and after thresholding respectively—so the mixed chemical character of the slow sample does not depend on where the velocity threshold is placed. That robustness, rather than any

individual fraction, is the result we draw from this subset. Per-star population-prior probabilities ship with the catalogue, so these regions can be recut at any threshold. This α -classified subset is dominated by luminous giants in the APOGEE and GALAH footprints; it is therefore a selective but useful chemical window, not a chemically complete census. The APOGEE [Mg/Fe] and GALAH [α /Fe] scales are not homogenised onto a single abundance system in this release, so the pooled classification is used only as a literature-region context flag. Survey counts are reported separately, and no whole-catalogue population fraction is inferred from the mixed-survey subset.

Within the 117-star α -classified sample, 51 stars fall in a Splash-like region, 23 in a Gaia-Sausage/Enceladus-like region, 22 in an Aurora-like region, 2 in a disc-like contaminant region, and 19 are not assigned to these literature regions. Weighting each star by its population-prior probability gives 26.1, 10.5, 11.4, 1.4, and 9.0 respectively; thresholding at $P > 0.5$ gives 22, 9, 10, 1, and 6. These counts are reported only within the α -classified sample. They are not used as whole-catalogue population fractions, because the remaining Tier A+B+C stars lack the high-resolution abundance information required for that assignment in the current release products.

The broader metallicity context is still informative. Gaia GSP-Phot metallicities are available for many candidates, but the comparison in Section 3.5 shows that GSP-Phot is biased toward solar metallicity for the cool, metal-poor giants that dominate this sample. We therefore use GSP-Phot only as a low-resolution heterogeneity check and rely on spectroscopic abundances for subset-qualified population language. Photometric metallicity alone does not supply the alpha-element information needed for the Splash/GSE/Aurora comparison cuts.

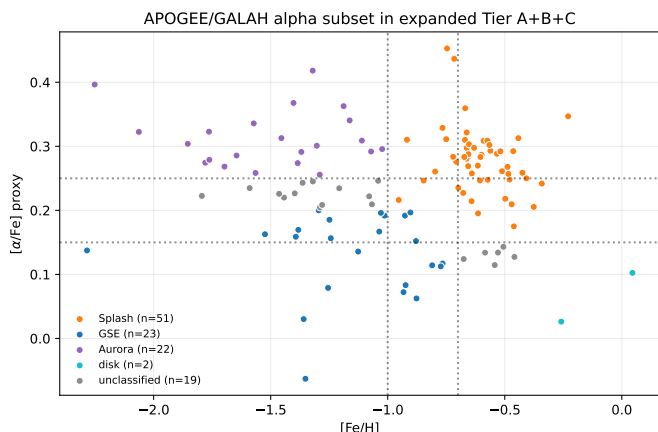


Fig. 10. [Mg/Fe] (APOGEE) and [α /Fe] (GALAH) vs. [Fe/H] for the α -classified spectroscopic subset within Tier A+B+C ($N = 117$). Literature-region labels are shown only as contextual comparisons and are not treated as whole-catalogue population assignments.

Fig. 10 shows the α -[Fe/H] plane for this subset, and Table 15 lists the corresponding literature-region decomposition; both are contextual diagnostics rather than whole-catalogue population assignments.

Table 14. High-resolution spectroscopic chemistry available for the Tier A+B+C catalogue. APOGEE DR17 (Abdurro'uf et al. 2022) and GALAH DR3 (Buder et al. 2021) provide the subset used for α -based literature-region comparisons. Matches come from a full exact-source-id VizieR TAP cross-match against the 20,829-source candidate pool. Gaia GSP-Phot metallicities are discussed only as a biased low-resolution contextual estimator in the text, not as population assignments.

Source	N	Median [Fe/H]
APOGEE DR17, usable α	67	-0.80
GALAH DR3, usable α	62	-0.83
unique stars after deduplication	117	-0.81
Splash classified	51	-0.62
GSE classified	23	-1.04
Aurora classified	22	-1.43
disc-like contaminant	2	-0.11
unclassified in these regions	19	-1.21

Table 15. Probability-weighted chemodynamic context. Spectroscopic abundances are rare in this catalogue, so the chemical discussion uses the full α -classified sample—every star with both [Fe/H] and an α proxy from APOGEE or GALAH ($n = 117$)—rather than only those above the adopted velocity threshold. Regions follow Belokurov et al. (2020), Helmi et al. (2018), and Belokurov & Kravtsov (2022). ΣP_{pop} is the summed population-prior membership probability, i.e. the effective number of genuinely slow stars each region contributes; $N(P > 0.5)$ is the count surviving the adopted threshold. The three columns give consistent proportions, so the mixed chemical character of the sample does not depend on where the velocity threshold is placed. Fractions are raw N shares with Beta($k + \frac{1}{2}, n - k + \frac{1}{2}$) Jeffreys 68% intervals (Cameron 2011); they are wide, which is itself an argument against extrapolating to the full catalogue. Per-star probabilities ship with the catalogue so these regions can be recut.

Population	N	ΣP_{pop}	$N(P > 0.5)$	Fraction	[Fe/H]
Splash	51	26.1	22	43.6 ^{+4.6} _{-3.9} %	-0.62
GSE	23	10.5	9	19.7 ^{+3.5} _{-3.4} %	-1.04
Aurora	22	11.4	10	18.8 ^{+3.8} _{-3.3} %	-1.43
Disc-like	2	1.4	1	1.7 ^{+1.6} _{-0.8} %	-0.11
Unclassified	19	9.0	6	16.2 ^{+3.7} _{-3.1} %	-1.21
α -classified	117	58.5	48	100.0%	—

3.7. Sky Distribution, Observer Geometry, and Selection

The sample is not isotropically distributed on the sky (Fig. 12). In Galactic coordinates, $39.7 \pm 1.1\%$ of the Tier A+B+C stars lie toward the Galactic-centre direction ($l \in [330^\circ, 30^\circ]$), while only $6.2^{+0.6}_{-0.5}\%$ lie toward the anti-centre direction ($l \in [150^\circ, 210^\circ]$). Latitude coverage is broad, from about -85° to $+87^\circ$, with a slight southern bias in the median latitude.

This concentration is a prominent descriptive feature of the observed catalogue. It likely reflects some combination of heliocentric observing geometry, the Gaia radial-velocity selection function, extinction and crowding effects, and the radial character of the underlying trajectories. The velocity-stratified controls are also centre-weighted, indicating that the sky distribution is partly a generic property of observed Gaia 6D stars at these distances.

The Castro-Ginard et al. (2023) Gaia DR3 RVS selection function (extending the EDR3 RVS sample-completeness framework of Overall & Boubert 2022) is therefore used as a limitation and contextual observability diagnostic for the observed catalogue rather than as a precise deprojection of the underlying Milky Way population. We evaluate the catalogue with the GaiaUnlimited DR3-RVS implementation and release the per-star selection-function table, but inverse-selection weighted counts are not used as primary population counts.

Table 16. Gaia DR3 RVS selection-function diagnostics for the catalogue. The catalogue is selected from the observed Gaia DR3 6D/RVS subset and is therefore not a volume-complete Milky Way census. N_{valid} is the number of sources for which the GaiaUnlimited DR3-RVS implementation (Castro-Ginard et al. 2023) returned a finite selection-function weight at the source’s $(G, BP-RP, l, b)$ cell; the $N - N_{\text{valid}}$ deficit is dominated by cells with zero parent count. $f_{n<10}$ is the fraction of N_{valid} sources whose underlying GaiaUnlimited cell carries a parent count below 10 (the regime in which the $\text{Beta}(k+1, n-k+1)$ posterior remains competitive with its $\text{Beta}(1, 1)$ prior; posterior standard error $\gtrsim 0.16$ for $n < 10$); $f_{\Sigma, n<10}$ is the share of $\Sigma 1/S_{\text{RVS}}$ contributed by those prior-dominated cells. Tiers are the adopted population-prior ones. The lower block restricts each tier to the inner-Galaxy quadrant $l \in [330^\circ, 30^\circ]$ where the catalogue concentrates (Section 3.7). Inverse-selection weighted sums are reported only as contextual observability diagnostics and are not used as primary population counts.

Sample	N	N_{val}	$\Sigma 1/S$	$\bar{S}$	$f_{n<10}$	$f_{\Sigma, n<10}$
<i>Full sample</i>						
Candidate pool	20,829	19,999	37,256.3	0.50	–	–
Tier A	173	172	273.5	0.67	84.3%	89.6%
Tier B	103	101	163.7	0.60	76.2%	83.9%
Tier C	345	343	584.7	0.50	84.5%	90.0%
Tier A+B	276	273	437.2	0.67	81.3%	87.5%
Tier A+B+C	621	616	1,021.9	0.67	83.1%	88.9%
<i>Inner-Galaxy quadrant, $l \in [330^\circ, 30^\circ]$</i>						
Tier A (inner)	17	17	22.1	0.90	52.9%	60.7%
Tier A+B (inner)	32	31	43.5	0.83	54.8%	63.5%
Tier A+B+C (inner)	88	87	138.6	0.67	67.8%	76.3%

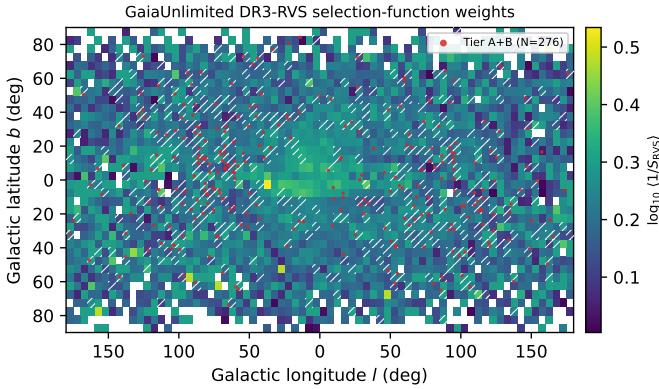


Fig. 11. GaiaUnlimited DR3-RVS (Castro-Ginard et al. 2023) selection-function weight map for the expanded candidate pool, queried in the same $(G_{\text{RVS}}, BP-RP, l, b)$ convention as the pipeline. Cell colour is the per-pixel mean $\log_{10}(1/S_{\text{RVS}})$; red points mark the Tier A+B primary catalogue ($N = 276$). White hatching flags pixels in which $\geq 50\%$ of catalogue stars sit in GaiaUnlimited cells with parent count $n < 10$, the regime where the $\text{Beta}(k+1, n-k+1)$ posterior is prior-dominated. The Tier A+B locus overlaps the hatched region substantially, consistent with the $f_{n<10} = 81.3\%$ of stars and 87.5% of $\Sigma 1/S_{\text{RVS}}$ reported in Table 16.

4. Orbit-Derived Properties in the Adopted Potential

Within the adopted static Hunter+2024 potential, the Tier A+B+C catalogue maps onto strongly radially biased model orbits, and this result is robust across a suite of diagnostic tests. The following subsections present the primary orbit-derived properties, compare them against velocity-stratified controls, describe how the sample’s properties vary with orbital scale, and assess robustness to measurement uncertainty and potential choice.

Table 17 identifies the sampling scheme behind the repeated cylindrical-pericentre and spin-axis-reach statistics used below. For the catalogue, the primary orbit summary uses 5,000-realisation orbit Monte Carlo propagation for every Tier A+B+C star in the adopted static Hunter+2024 potential. The velocity-stratified control comparison, halo-mass variants, barred potential variants, and cylindrical spin-axis-reach counts are retained as explicitly labelled point-estimate diagnostics.

4.1. In the Adopted Model, the Sample Is Strongly Radially Biased

Within the adopted static Hunter+2024 potential, the orbit-derived picture is strongly radial. For the Tier A+B+C catalogue ($N = 621$), the median of the per-star posterior-median eccentricities is 0.965. All 621 stars have $e > 0.8$, $72.6 \pm 1.8\%$ have a posterior-median $e > 0.95$, and $1.9^{+0.6}_{-0.5}\%$ have a posterior-median $e > 0.99$ from the 5,000-realisation Monte Carlo orbit product (Table 18; Jeffreys 68% intervals).

The corresponding fractions obtained from a single deterministic point-estimate orbit integration in the same potential (Table A.2) are higher: $66.7 \pm 1.1\%$ of Tier A+B+C stars have $e > 0.95$ under point-estimate integration. The difference between the Monte Carlo posterior-median fraction (48.2%) and the point-estimate fraction (66.7%) is the expected consequence of convolving the measurement-error distribution into a quantity that is bounded above by unity: the per-star posterior medians regress slightly toward lower e relative to the noise-free point estimate. Both numbers describe the same catalogue; we report the posterior-median fraction as the primary value because it propagates the observational uncertainties explicitly.

The same Monte Carlo orbit product gives a Tier A+B+C median cylindrical R_{peri} of 154 pc and a median cylindrical R_{apo} of 7.35 kpc. The maximum posterior median R_{apo} in the present Tier A+B+C integrations is 17.86 kpc. These values confirm that the low-speed catalogue is not dominated by ordinary near-circular disc kinematics. This compact, radial mapping is also physically expected for a catalogue selected at very low total Galactic rest-frame speed: in the adopted Galactocentric frame, such stars carry little angular momentum at the observed epoch. The qualitative compact-pericentre, high-radiality tendency is therefore robust across the tested potentials; the precise per-star cylindrical pericentres are not, because the real inner Galaxy is barred and non-axisymmetric.

The same integrations settle what the orbits do at closest approach, which a small pericentre alone does not determine. The azimuth swept between successive apocentres is $2\pi \Omega_\phi / \Omega_R$: it is 360° for a closed Kepler ellipse, in which the star returns along its inbound track, and 180° for a harmonic, constant-density core, in which successive apocentres lie on opposite sides of the centre. For the Tier A+B+C catalogue the median is 195° , with a 16th–84th percentile range of 155° – 202° , and no member exceeds 300° ; the median is the same to within 3° in each individual tier. These are therefore transiting rather than reversing orbits: the star crosses the inner Galaxy and emerges on the far side, and its pericentre is a rapid passage rather than a turning point at which the motion is undone. The proximity of the median to the harmonic value is consistent with orbits that spend their perigalactic phase inside the region where the rotation curve is still rising. The frequencies are Stäckel-fudge outputs and inherit the accuracy caveats of Appendix A, so we read the distribution as a whole rather than individual per-star

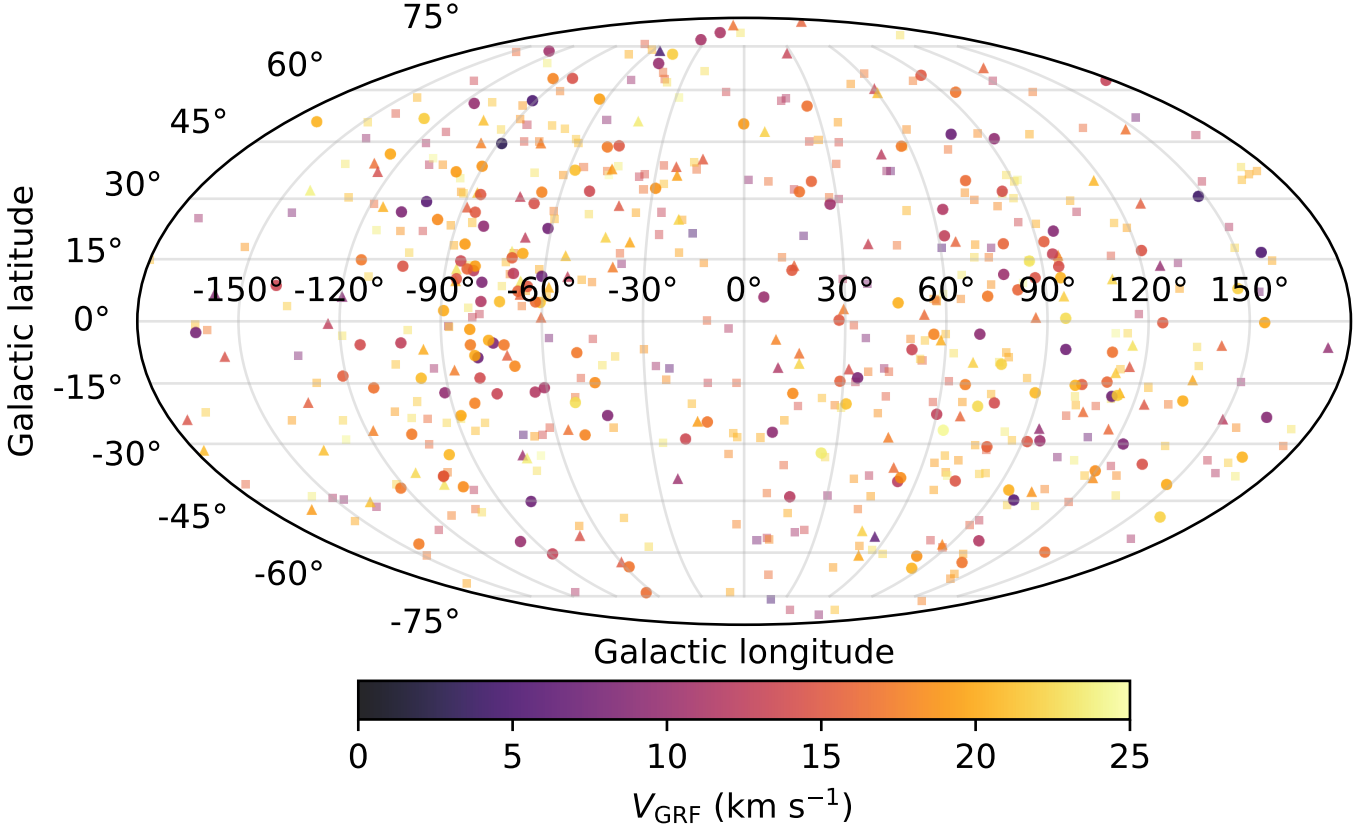


Fig. 12. Aitoff projection of the Tier A+B+C slow catalogue ($N = 621$) in Galactic coordinates ($l = 0$ centred, l increasing to the left), colour-coded by V_{grf} . Marker shape encodes tier (circle/triangle/square = A/B/C). The catalogue concentrates strongly toward the inner Galaxy.

Table 17. Ledger for repeated cylindrical-pericentre and spin-axis-reach statistics. Each row refers to a different sampling scheme or sensitivity test, so the numerical values are not interchangeable. The first row is the Monte Carlo posterior-median orbit product; the control and sensitivity rows are point-estimate diagnostics unless stated otherwise.

Use in paper	Sampling scheme	Quantity reported	Interpretation
Primary orbit summary	5,000-realisation orbit Monte Carlo in the static Hunter+2024 potential	Median $R_{\text{peri}} = 154$ pc, median $R_{\text{apo}} = 7.35$ kpc, median $e = 0.949$	Uncertainty-propagated cylindrical-orbit summary used for Table 18.
Control comparison	Slow-catalogue point estimate; existing controls re-weighted to catalogue observability marginals	Slow-row point-estimate medians: $R_{\text{peri}} = 115.5$ pc and $e = 0.964$; 200–260 km s^{-1} $R_{\text{peri}} = 2.20$ kpc	Supports the endpoint interpretation in Table 19.
Cylindrical spin-axis reach	Point-estimate static and barred integrations	Static: 241 stars with $R_{\text{peri}} < 100$ pc; barred: 526	Shows strong cylindrical inner reach; it is not a count of close approaches to Sgr A*.
Halo/bar/distance sensitivity	Point-estimate halo and bar-speed reruns; BJ/parallax coupling stress test	Halo variants: median $R_{\text{peri}} = 111$ –119 pc; bar variants: 7–31 long-apocentre inner-reach candidates	Shows the main radial-orbit result is stable while inner-Galaxy reach counts remain potential-sensitive.

values; the conclusion rests on its distance from 360° , which is far larger than the fudge error.

These quantities should be read as outputs of the adopted model rather than as direct Gaia observables. They are useful because they provide a coherent dynamical interpretation of the slow state. Cylindrical pericentre estimates below roughly 200 pc should be treated as indicative, because the real inner Galaxy is barred and non-axisymmetric on those scales; the barred Hunter/Sormani variant in Section 4.5 brackets this sensitivity. Satellite-orbit studies likewise find that pericentres are often more stable than orbit type or apocentre under large-scale potential changes (Li et al. 2021; Pace et al. 2022), but those satellite-scale results are not used here as a quantitative error model for sub-kpc inner-Galaxy reaches. They instead motivate the emphasis on population distributions, matched-control trends, and threshold counts rather than exact per-star cylindrical turning points or three-dimensional closest approaches.

Crucially, the orbital *shape* is robust to the choice of axisymmetric potential. Re-integrating the catalogue in the independently calibrated, Gaia-DR3-external McMillan (2017) potential gives a near-identical point-estimate summary—median eccentricity 0.968, median $R_{\text{peri}} = 138$ pc, median $R_{\text{apo}} = 8.24$ kpc, all 621 stars with $e > 0.8$, and three static long-apocentre inner-reach candidates—so the radial, compact-pericentre character is not an artefact of the non-Gaia-DR3-calibrated Hunter+2024 model (Appendix A). What is potential-sensitive is the precise cylindrical spin-axis reach and the barred long-apocentre inner-reach count, which vary by a factor of several between the static and barred families (§4.4; Appendix A). We therefore treat the high-eccentricity, small- R_{peri} summary as robust and the exact inner-Galaxy reach as model-dependent.

Table 18. Summary of orbit-derived properties for the Tier A+B+C catalogue ($N = 621$) under 5,000-realisation Monte Carlo orbit propagation in the Hunter+2024 axisymmetric Galactic potential. Reported values are summaries of the per-star posterior medians; R_{peri} and R_{apo} are cylindrical radial turning points, while z_{max} is the maximum vertical excursion, computed over a 4 Gyr forward integration. Eccentricity-fraction uncertainties are Beta($k + \frac{1}{2}, n - k + \frac{1}{2}$) Jeffreys 68% intervals (Cameron 2011); the corresponding point-estimate fractions are in Table A.2.

Quantity	Value
Median posterior-median e	0.949
Posterior-median $e > 0.8$ (%)	100.0
Posterior-median $e > 0.95$ (%)	48.2 ± 1.1
Posterior-median $e > 0.99$ (%)	$0.7^{+0.2}_{-0.2}$
Median R_{peri} (pc)	154
R_{peri} p16 (pc)	89
R_{peri} p84 (pc)	274
Median R_{apo} (kpc)	7.35
Maximum R_{apo} (kpc)	17.86
Median z_{max} (kpc)	3.35

Fig. 14 places the sample in the Toomre diagram: the catalogue concentrates at low azimuthal velocity, far from the disc locus, as the low- V_{grf} selection requires.

4.2. Matched-Control Comparison

A velocity-stratified, observability-matched control comparison provides the clearest test of whether the slow-star compact-pericentre and high-eccentricity orbit summaries merely reflect

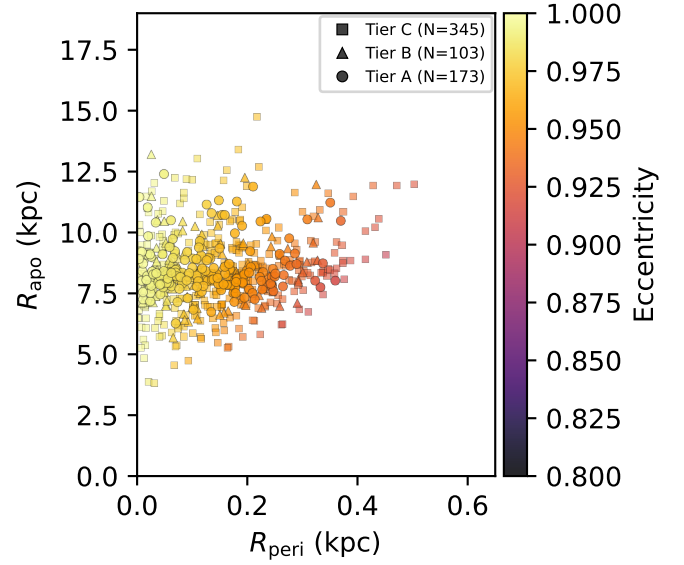


Fig. 13. Cylindrical pericentre vs. cylindrical apocentre for the Tier A+B+C slow catalogue ($N = 621$), colour-coded by eccentricity, in the static Hunter+2024 potential. Axis ranges are set to where the adopted-model point-estimate integrations place the data: all 621 stars have $R_{\text{peri}} < 1$ kpc and nearly all have $R_{\text{apo}} < 18$ kpc. The long-apocentre inner-reach sensitivity test (§4.4) is reported in the text and in Table 21, not on this panel.

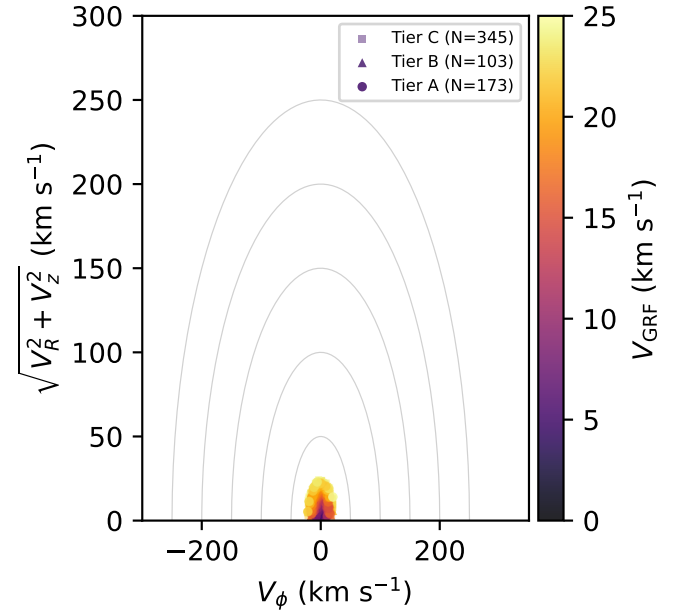


Fig. 14. Toomre diagram (V_ϕ vs. $\sqrt{V_R^2 + V_z^2}$) for the Tier A+B+C slow catalogue ($N = 621$), colour-coded by V_{grf} . The sample is concentrated at low azimuthal velocity, far from the disc locus.

Gaia stars at similar distances, or whether they vary systematically with Galactic rest-frame speed. The controls are weighted toward the slow catalogue’s broad observability marginals, and the covariate balance of that weighting is audited in Section 2.5. In the catalogue, the slow-sample point-estimate median R_{peri} is 115.5 pc. The control-band median pericentres rise monotonically with V_{grf} under every weighting scheme; the exact values are method-dependent but the ordering and the slow sample’s compact-pericentre endpoint are not. Under the covariate-balanced entropy weights (Table 6) the four control medians are 206 pc, 592 pc, 2.68 kpc, and 3.97 kpc; under the observability kernel they are 132.6 pc, 408 pc, 1.36 kpc, and 2.25 kpc; and unweighted they are 143 pc, 460 pc, 1.76 kpc, and 2.82 kpc. In every case the slow catalogue sits at the compact-pericentre extreme of the matched controls. We read this matched comparison as a null check that the slow sample’s compact pericentres are not generic to Gaia 6D stars at similar observability, not as a calibrated speed–pericentre relation: the controls are a deliberately broad, selection-function-limited comparison over four coarse velocity bands, and the across-band gradient runs in the direction expected from the kinematics of the cut.

The same comparison provides a direct check on bar-resonance accumulation. The slow sample is not uniquely enhanced at the OLR: its weighted OLR fraction is 1.84%, lower than the 100–200 and 200–260 km s^{-1} controls and only modestly higher than the adjacent low-velocity controls. Barred integrations therefore remain important for individual inner-Galaxy orbits and for the long-apocentre inner-reach sensitivity test, but the matched-control comparison does not support OLR accumulation as the primary explanation for the catalogue. Figure C.1 in Section 4.6 places the catalogue in L_z – J_R action space alongside the inner Lindblad, corotation, and outer Lindblad loci for the full Ω_p grid; the Tier A+B+C distribution concentrates near $L_z \approx 0$ rather than at any single pattern-speed-resonance pile-up.

The full eccentricity CDFs reinforce the same point. An unweighted two-sample Anderson–Darling comparison (Scholz & Stephens 1987) between the Tier A+B+C static-potential eccentricities and the adjacent 25–50 km s^{-1} matched-control CDF rejects the hypothesis of a common eccentricity distribution at asymptotic $p < 0.001$. The same test against an eccentricity realisation drawn from a single trivariate-Gaussian velocity ellipsoid fitted to the control library likewise rejects at asymptotic $p < 0.001$ (the mock realisation and the script that regenerates it from the released control library are included in the data bundle). Thus the near-unity eccentricity concentration is not reproduced by either the adjacent observed control band or that smooth velocity ellipsoid. Both are statements about the *shape* of the eccentricity distribution and require no parent normalisation, which is why they survive the considerations of Section 3.3. The monotonic eccentricity decline across the four control bands is likewise recovered under the covariate-balanced control weights. Fig. 15 shows the weighted per-band median eccentricities (with the full per-band distributions in Fig. 16), and Fig. 17 the corresponding per-band pericentre medians.

Table 19. Matched-control comparison across V_{grf} velocity bands. Each band is integrated in the Hunter+2024 axisymmetric potential. The R_{peri} and e columns are deterministic point-estimate orbit summaries; the Monte Carlo posterior-median eccentricity summary is given separately in Table 18. Control rows use entropy-balanced weights that match the slow catalogue’s observability covariate means (Table 6); the final column gives the older product-kernel cylindrical-pericentre median only as a sensitivity readout.

Sample	N	N_{eff}	R_{peri}	e	$R_{\text{peri,kern}}$
slow A+B+C	621	—	132	0.970	—
25–50	1,955	387	206	0.921	133
50–100	2,586	990	592	0.796	408
100–200	2,788	1,438	2,676	0.390	1,362
200–260	2,749	1,474	3,974	0.237	2,246

Control-row labels are V_{grf} bands in km s^{-1} ; R_{peri} is the cylindrical pericentre in pc and e is dimensionless. The slow row uses the parent-complete Tier A+B+C catalogue and the point-estimate orbit product, so its $e = 0.964$ should not be interchanged with the MC posterior-median $e = 0.949$ in Table 18. Entropy-balanced effective sample sizes are rounded to the nearest star.

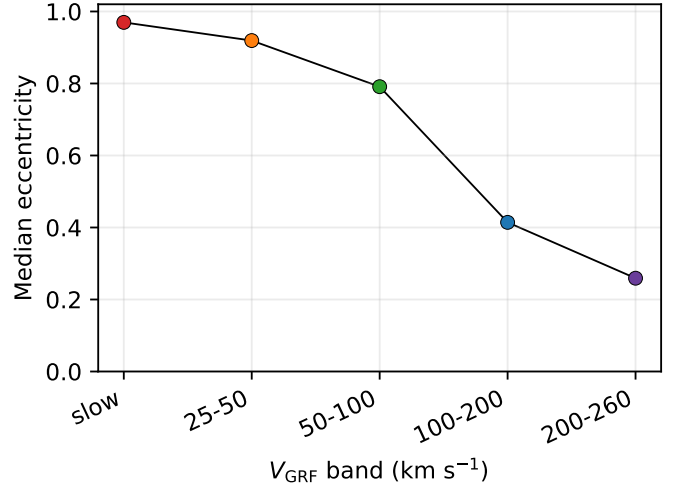


Fig. 15. Median eccentricity as a function of Galactic rest-frame velocity for the Tier A+B+C slow catalogue ($N = 621$) and the four observability-matched control bands. Points show the band median eccentricities after the control bands are weighted against the slow catalogue; the full per-band eccentricity distributions are shown in Fig. 16. The monotonic decline from $e \sim 1.0$ at low V_{grf} to moderate eccentricity at 200–260 km s^{-1} matches the band-by-band medians in Table 19.

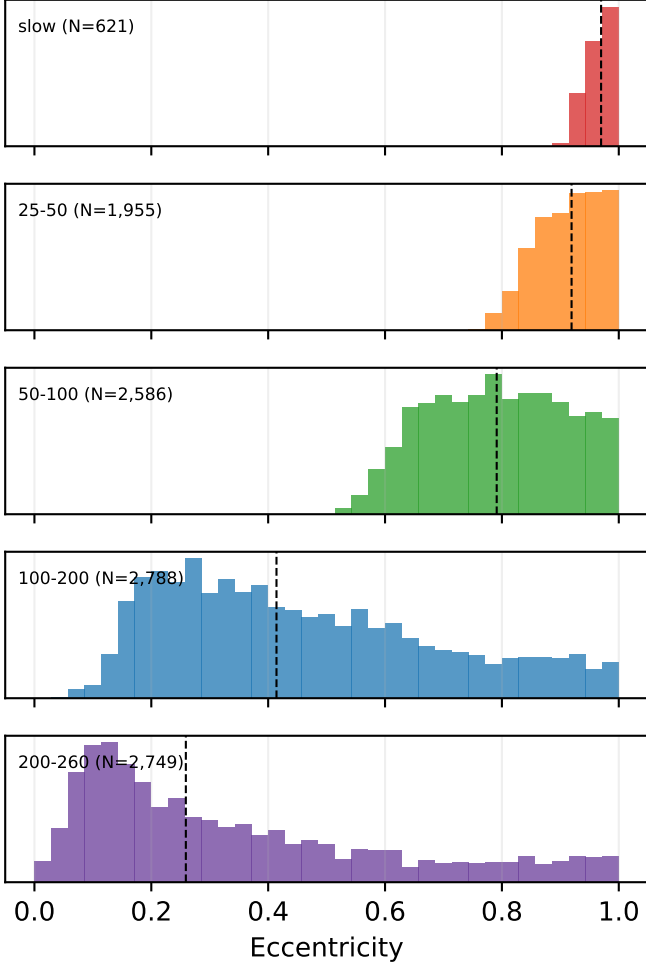


Fig. 16. Eccentricity histograms for each of the five GRF velocity bands, stacked vertically. The slow Tier A+B+C band ($N = 621$) shows the extreme near-unity peak; the eccentricity distribution broadens monotonically through the four control bands toward 200–260 km s^{-1} .

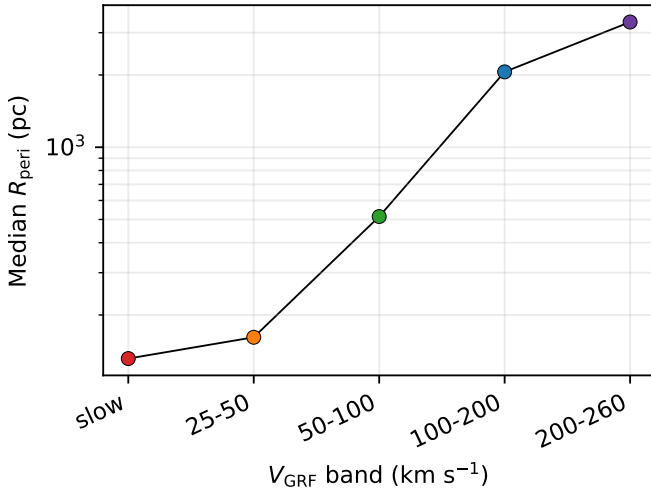


Fig. 17. Median cylindrical pericentre as a function of Galactic rest-frame velocity for the slow Tier A+B+C catalogue ($N = 621$) and the four observability-matched control bands. The slow band sits at the small- R_{peri} extreme of the matched controls; per-band pericentre medians are reported in Table 19.

4.3. Dwell-Time Analysis in the Adopted Potential

A dwell-time calculation in the adopted static Hunter+2024 potential helps quantify what the low- V_{grf} selection captures (Table 20). Tier A stars spend a median 28.3% of their integrated orbital time inside $R < 2$ kpc, while Tier C stars spend 19.7% there. Conversely, the median time fraction in the solar annulus ($7 < R < 9$ kpc) rises from 11.6% in Tier A to 16.2% in Tier C. The high-confidence slow stars are therefore not merely objects lingering in a solar-radius comparison shell; they are on strongly radial orbits that cross the inner Galaxy, observed during a low-speed phase.

Table 20. Tier-resolved orbital dwell statistics in the Hunter+2024 potential. Reported are the per-tier medians of the time fraction spent within selected cylindrical radius shells over a 4 Gyr integration, indicating that high-confidence slow- V_{grf} stars spend a disproportionate fraction of their trajectory inside $R < 2$ kpc despite being observed at the solar circle.

Shell	Tier A	Tier B	Tier C
Median $f(R < 1 \text{ kpc})$ (%)	14.2	11.8	9.4
Median $f(R < 2 \text{ kpc})$ (%)	28.3	24.1	19.7
Median $f(R < 5 \text{ kpc})$ (%)	65.4	60.8	55.0
Median $f(7 < R < 9 \text{ kpc})$ (%)	11.6	13.5	16.2

4.4. Exploratory Inner-Galaxy Reach and Selection Limits

Although the catalogue shares the defining low- V_{grf} state, the stars span a range of model-derived orbital scales. The Tier A+B+C catalogue has a median static-potential cylindrical pericentre of 153 pc in the orbit Monte Carlo posterior medians, and all 621 posterior-median cylindrical pericentres remain inside 1 kpc. In the dedicated point-estimate cylindrical-reach diagnostic, R_{peri} is the cylindrical in-plane pericentre, $\min \sqrt{x^2 + y^2}$, i.e. closest approach to the Galactic spin axis rather than to Sgr A* as a point. With this definition, 241 stars (38.8%) have static $R_{\text{peri}} < 100$ pc and all 621 have static $R_{\text{peri}} < 1$ kpc. This fact is visible in the right panel of Fig. 6: every star's model cylindrical pericentre falls within the inner kpc in R , despite the catalogue being observed across the Gaia DR3 heliocentric survey volume shown in the left panel. In the barred sensitivity integration the corresponding cylindrical point-estimate count is 1,635 stars with $R_{\text{peri}} < 100$ pc; those barred quantities are used as sensitivity diagnostics rather than as the default catalogue values. The conservative cylindrical inner-reach statement is therefore stronger than the exact 100-pc scale: in the adopted static point-estimate model, the full catalogue lies well inside common bar/bulge apertures ($R < 3.5$ – 5 kpc), while the parsec-scale cylindrical medians and counts remain model diagnostics.

We additionally propagated dedicated 5,000-realisation Monte Carlo closest approaches to the Galactic centre for the closest static and barred seeds: no candidate has $P(r_{\text{sph},\text{min}} < 10 \text{ pc}) > 0.5$ and six have $P(r_{\text{sph},\text{min}} < 100 \text{ pc}) > 0.5$, so we describe these as candidate inner-Galaxy approachers rather than a confirmed Sgr A*-passing population. The closest-approach probabilities and cumulative curves are reported in Appendix B (Table B.1, Fig. B.1).

The closest-approach calculation follows the same general philosophy as Gaia DR3 stellar-encounter work, where orbit integration and uncertainty propagation are used to identify robust close-passage candidates rather than to treat nominal closest approaches as definitive (Bailer-Jones 2022).

As an external source-id check, none of the 276 Tier A+B stars overlaps the 61 Gaia DR3 close-encounter entries in Bailer-Jones (2022).

We also checked the closest published low-azimuthal-velocity comparison sample, the 69 slow- and retrograde APOGEE/Gaia stars of Filion et al. (2025). All 69 resolve to unique Gaia DR3 sources and pass the present Gaia-RVS parent-source contract, but none enters Tier A+B+C under the present low- V_{grf} probability tiering. Three enter the 20,829-source Monte Carlo candidate pool and remain Tier X. The detailed comparison is reported in Appendix F (Table F.1); it confirms that the difference is primarily one of selection definition—low V_{ϕ} is not the same as low total V_{grf} —rather than an exact source-id overlap.

The radial-reach tail is also potential-sensitive. Three Tier A+B+C stars (Poisson 68% band 1.4–5.9) satisfy the long-apocentre inner-reach criterion ($R_{\text{peri}} < 2$ kpc and $R_{\text{apo}} > 15$ kpc) in the static Hunter+2024 integration. The barred default integration produces 21 such candidates (68% Poisson band 16.5–26.7); across the pattern-speed grid explored in Section 4.5, the count ranges from 7 to 31. Because the absolute count varies by a factor of several between the static and default-barred integrations and remains Ω_p -sensitive within the barred family, we treat these stars as bar-sensitive orbital-geometry candidates whose count is intrinsically potential-dependent, rather than as a robust population count or a merger-debris census.

Table 21. Inner-galaxy reach of the Tier A+B+C catalogue ($N = 621$). Counts are 40,001-point, parabola-interpolated point-estimate diagnostics for stars with small cylindrical in-plane pericentre $R_{\text{peri}} = \min \sqrt{x^2 + y^2}$, very small minimum spherical Galactocentric distance r_{sph} , and "extreme reach" trajectories spanning the disc-to-halo radial range; the Monte Carlo orbit summary is reported separately in Table 18. The Static and Barred columns integrate under the Hunter+2024 static and barred potentials respectively; the Portail+2017 column uses the made-to-measure inner-Galaxy barred potential (Sormani-Vasiliev Agama recipe) rotated at $\Omega_p = 37.5 \text{ km s}^{-1} \text{ kpc}^{-1}$ to bracket the long-apocentre inner-reach count against an independent gravitational background. Candidate-row uncertainties are Gehrels (1986) Poisson 68% intervals. The barred median pericentre readout remains sampling-sensitive at tens-of-pc scales, so threshold counts are the more stable barred inner-reach diagnostic.

Diagnostic	Static	Hunter+24	Portail+17
$R_{\text{peri}} < 100 \text{ pc}$	241	526	526
$R_{\text{peri}} < 50 \text{ pc}$	130	486	486
$R_{\text{peri}} < 10 \text{ pc}$	24	145	145
Min $r_{\text{sph}} < 100 \text{ pc}$	113	208	208
Min $r_{\text{sph}} < 10 \text{ pc}$	1	1	1
Long-apocentre reach	$3^{+2.9}_{-1.6}$	$21^{+5.7}_{-4.5}$	$15^{+5.0}_{-3.8}$

4.5. Robustness and Sensitivity

The principal dynamical claims were tested against quality filtering, threshold variation, measurement uncertainty, distance estimator, Solar frame, halo mass, and barred-potential sensitivity. The full battery is collected in Appendix A: across the Gold-subset, threshold, Monte Carlo, halo-mass, barred-potential, and Portail+2017 tests, the finite-time-Lyapunov chaos diagnostic, and the Stäckel-fudge action-accuracy audit, the qualitative result is unchanged—the catalogue is compact and strongly radial, its inner-reach tail is bar-sensitive, and the high-eccentricity character survives every quality cut. Central

68% intervals on the primary numerical claims are collected in Table 22.

4.6. Energy and Angular-Momentum Distribution

A standard Gaia-era diagnostic is the orbital energy versus z -component of angular momentum ($E-L_z$); the same diagnostic underlies the substructure surveys of Naidu et al. (2020) and the proto-Galaxy mapping of Rix et al. (2022). In the static Hunter+2024 potential, the Tier A+B+C catalogue occupies low angular momentum and strongly bound inner-Galaxy phase space—the near-zero- L_z regime probed by earlier solar-neighbourhood studies (Hunt et al. 2016). Hunt et al. reported a dearth of disc stars near zero angular momentum, consistent with strong nuclear scattering of low- L_z orbits; that result is not a contradiction of the present sample because our selection is not a disc-only solar-neighbourhood distribution, but it is a useful caution that compact-pericentre low- L_z orbits should not be read as long-lived regular disc orbits. This supports the same catalogue-first interpretation used throughout the paper: the sample overlaps familiar Gaia-era accretion and in-situ regions, but the present work does not assign progenitor membership from $E-L_z$ alone. The per-star catalogue reports E , L_z , and Stäckel-fudge actions for follow-up clustering work.

The same L_z axis (Fig. 18) also locates the catalogue relative to the bar resonances. In L_z-J_R Stäckel-fudge action space the Tier A points concentrate near $L_z \approx 0$ at large J_R , away from any single corotation or outer-Lindblad pile-up, consistent with the matched-control OLR-fraction check in Section 4.2. Because the Stäckel-fudge vertical actions carry a large uncertainty for these plunging orbits (full-sample audit: median $|\Delta J_z| \approx 85\%$ in Tier A and 62% over Tier A+B+C, versus $\lesssim 10\%$ for J_R and L_z ; Appendix A), we read this overlay as qualitative context rather than a quantitative resonance classification, in the spirit of recent Gaia DR3 action-space work (Palicio et al. 2023). The overlay is shown in Appendix C (Fig. C.1).

Table 22. Uncertainty ledger for the primary numerical claims. Intervals are central 68% ranges: bootstrap resampling of the catalogue for the medians and observed-control ratios, Wilson score intervals for the binomial pipeline fractions, and score-implied Bernoulli standard errors for the purity rows. Rows labelled “point” use deterministic point-estimate orbit products; R_{peri} and R_{apo} are cylindrical. The uncertainty-propagated eccentricity summary is Table 18.

quantity	value	68% interval	method
Median eccentricity (Tier A+B+C, point)	0.964	0.963–0.965	bootstrap
Median R_{peri} (pc, point)	115.5	112–119	bootstrap
Median R_{apo} (kpc, point)	7.35	7.27–7.47	bootstrap
Control 25–50 median R_{peri} (pc)	132.6	128–135	weighted bootstrap
Score-implied purity (Tier A+B, %)	94	93–95	score Bernoulli SE
Score-implied purity (Tier A+B+C, %)	73	72–74	score Bernoulli SE
Observed-control ratio (BIC mixture)	7.4	7.3–7.7	bootstrap
Observed-control ratio (single ellipsoid)	25.6	25.3–26.2	bootstrap
Pipeline recovery, conditional (%)	87.0	86.5–87.4	Wilson
Pipeline recovery, end-to-end (%)	60.4	59.9–60.9	Wilson
25–50 leakage fraction (%)	3.13	3.0–3.3	Wilson

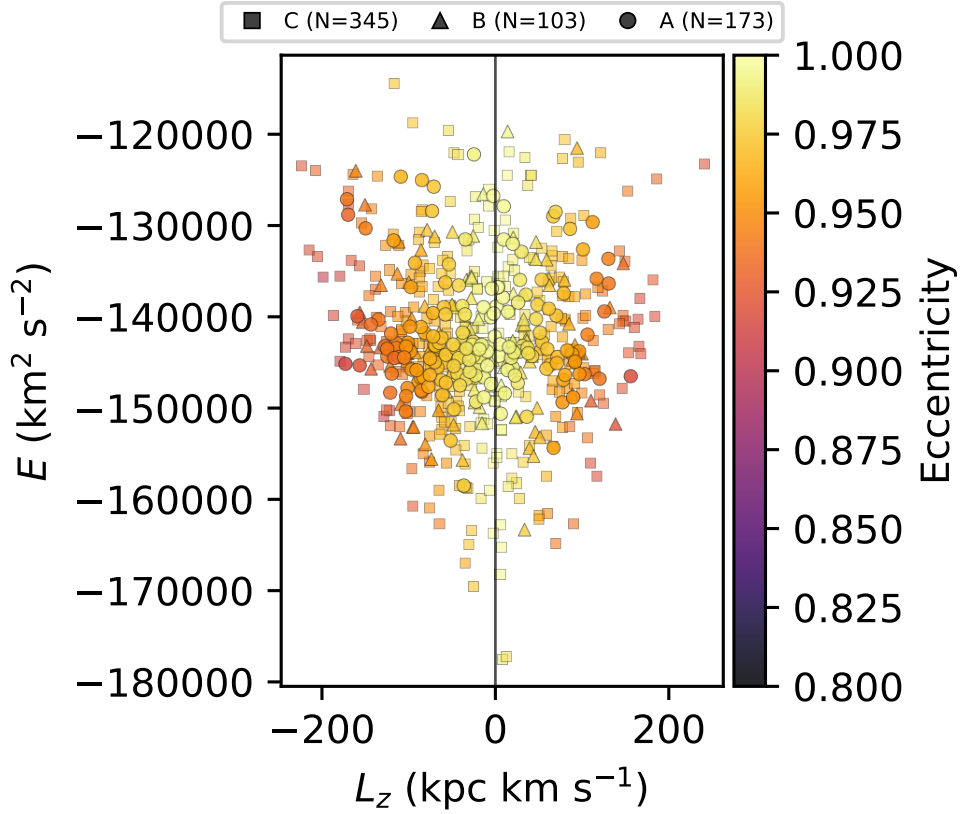


Fig. 18. Tier A+B+C slow catalogue ($N = 621$) in E – L_z space, colour-coded by orbital eccentricity in the static Hunter+2024 potential. Marker shape encodes tier (circle/triangle/square = A/B/C); the vertical line marks $L_z = 0$.

4.7. Pipeline Injection-Recovery

schedule described above. For consistency with the v2 selection-function audit, the RVS-availability draw preserves the v1 GaiaUnlimited convention $g = G_{\text{RVS}}$ and $c = BP - RP$. Table 23 and Fig. 19 show that 6,038 of 10,000 true $0-25 \text{ km s}^{-1}$ injections are recovered as Tier A+B+C, while only 313 of 10,000 stars in the adjacent $25-50 \text{ km s}^{-1}$ decade leak into the slow catalogue and the higher-velocity leakage is negligible. Conditional on entering the Monte Carlo after the RVS and parallax-SNR filters, the slow-bin recovery is $6,038/6,944 = 87.0\%$; end-to-end, including Gaia/RVS observability, it is 60.4% . Combined with the latent-distribution reconstruction of Section 3.4 and the selection-function prior-dominance audit in Section 5, the injection-recovery test forms the catalogue's completeness triad: the latent V_{grf} distribution passes smoothly through the selection boundary, so the catalogue is a tail selection rather than a distinct population; its effective-volume scaling is limited by sparse selection-function cells; and the pipeline recovers a quantified fraction of true synthetic slow stars.

Table 23. GeDR3mock injection–recovery summary. Each true- V_{grf} bin contains controlled kinematic injections into GeDR3mock sightlines, photometry, stellar parameters, and Gaia-like uncertainties after the initial $G < 17$, $G_{\text{RVS}} < 14$, and $3500 < T_{\text{eff}} < 7000 \text{ K}$ observability cuts. N_{RVS} is the number passing a stochastic GaiaUnlimited DR3-RVS draw evaluated with the same v1 convention used for the catalogue, $g = G_{\text{RVS}}$ and $c = BP - RP$; N_{ϖ} is the number passing $\varpi_{\text{corr}}/\sigma_{\varpi} > 5$; N_{input} is the intersection entering the 500-realisation velocity-threshold Monte Carlo; and N_{rec} is the number recovered as Tier A, B, or C ($P(V_{\text{grf}} < 25 \text{ km s}^{-1}) > 0.5$).

True V_{grf} bin	N_{true}	N_{RVS}	N_{ϖ}	N_{input}	$N_{\text{rec}}/N_{\text{true}}$
$0-25 \text{ km s}^{-1}$	10,000	7,001	9,891	6,944	6,038/10,000 (0.604)
$25-50 \text{ km s}^{-1}$	10,000	7,035	9,896	6,972	313/10,000 (0.031)
$50-100 \text{ km s}^{-1}$	10,000	7,090	9,889	7,030	1/10,000 (0.0001)
$100-200 \text{ km s}^{-1}$	10,000	6,966	9,879	6,906	0/10,000 (0.000)

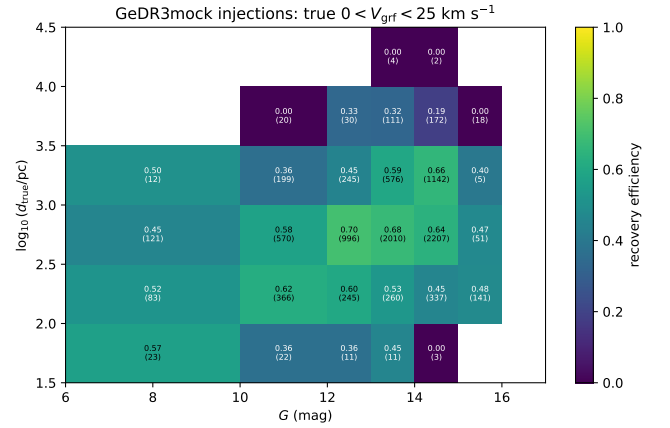


Fig. 19. End-to-end recovery efficiency for the true $0 < V_{\text{grf}} < 25 \text{ km s}^{-1}$ GeDR3mock injections as a function of apparent G magnitude and true distance. Each cell reports the recovered fraction, with the number of injected stars in parentheses. The denominator is the post-observability injected sample; the numerator requires the RVS-availability draw, $\varpi_{\text{corr}}/\sigma_{\varpi} > 5$, and Tier A+B+C recovery in the fixed 500-realisation diagnostic pass.

5. Interpretation and Limits

Empirically, Gaia DR3 contains a tiered catalogue of stars observed at very low Galactic rest-frame speed: 276 stars at

To measure the direct pipeline efficiency, we injected controlled kinematic realisations into GeDR3mock sightlines, photometry, stellar parameters, and Gaia-like uncertainties (Rybizki et al. 2020). The mock contains 10,000 stars in each true- V_{grf} interval $0-25$, $25-50$, $50-100$, and $100-200 \text{ km s}^{-1}$, after the initial $G < 17$, $G_{\text{RVS}} < 14$, and RVS-template-temperature cuts. Each injected star is passed through the same Lindegren zero-point, distance-posterior, $\varpi_{\text{corr}}/\sigma_{\varpi} > 5$, and Gaia DR3 RVS-availability stages used by the catalogue, plus a fixed 500-realisation velocity-threshold Monte Carlo diagnostic pass. The released catalogue itself uses the adaptive 500/5,000/10,000

Tier A+B confidence and 621 stars at Tier A+B+C confidence. The exact $V_{\text{grf}} = 25 \text{ km s}^{-1}$ boundary is less secure for individual near-threshold sources, so the catalogue is explicitly probabilistic rather than a single hard-cut list.

A velocity-stratified control comparison across four bands shows that the slow sample is more eccentric and compact-pericentre than matched controls at higher Galactic rest-frame speed, with the across-band gradient running in the expected kinematic direction. We read this as a null check that the slow sample's compact-pericentre character is not generic to Gaia DR3 6D stars at matched observability, not as a calibrated speed-pericentre relation: the broad, selection-function-limited control over four coarse bands is not adequate to establish such a law, and the gradient's direction is in any case largely set by the kinematics of the cut.

The dwell-time analysis (Section 4.3) and the matched-control gradient (Section 4.2) together make a single physical statement. A kinematic cut at very low V_{grf} selects stars on inner-reaching, radially biased orbits during a low-speed phase. This is related to the same orbital-turning-point mechanism discussed in the accretion-debris literature, but the present catalogue does not identify the selected stars with GS/E or any other progenitor. We emphasise that this strong radial bias is in part a kinematic corollary of the selection itself. In the standard spherical limit, spherical-radius pericentre and apocentre are the two turning-point roots of the radial energy equation, and eccentricity is set by those roots for given energy and angular momentum (Binney & Tremaine 2008; Hattori & Yoshii 2010, 2011). At fixed observed position, a star with $V_{\text{grf}} \rightarrow 0$ has little kinetic energy, so a bound orbit is being observed at, or very near, an outer turning point. Because the same limit also drives $|\mathbf{r} \times \mathbf{v}|$ toward zero, there is little centrifugal support to keep the next half-orbit near that same Galactocentric radius; in an ordinary centrally concentrated Galactic potential, the trajectory must move back toward smaller radii until it reaches its inner turning point. This generalizes the observability geometry noted by Hunt et al. (2024) in the context of disc radial phase spirals, where a fixed local volume preferentially catches low- L_z stars close to apocentre. High eccentricity and compact model cylindrical pericentres are therefore largely expected from the cut, wherever within the catalogue's observed Galactocentric range the slow state is measured. The model-dependent questions are the exact parsec-scale cylindrical spin-axis reach and, more stringently, the three-dimensional Galactocentric closest approach, not the direction of the radiality inference. The radial bias itself is thus largely a property of the selection; what the catalogue contributes is the slow sample, its spatial and orbital structure, and its count relative to smooth velocity-field continuations of the controls (§3.3).

The point-estimate orbit integration provides the current characterisation for the main dynamical claim. All 621 Tier A+B+C stars have $e > 0.8$ in the default static potential, and $74.9 \pm 1.7\%$ have $e > 0.95$ under point-estimate integration (cf. the Monte Carlo posterior-median fraction in Section 4.1). The strongly radial character of the sample is therefore clear in the adopted model, while posterior orbit uncertainties and potential-variant tests remain sensitivity products rather than catalogue-definition steps.

It is worth stating explicitly for how long these integrations carry information about a real star, because an orbit integrated for several gigayears increasingly describes the adopted potential rather than the object. The predictive content is front-loaded. A star's present position and velocity are measured, so the direction of its next excursion is an observable rather than

a model output: a star observed at $V_{\text{grf}} < 25 \text{ km s}^{-1}$ near an outer turning point is falling inward next, whichever centrally concentrated potential is assumed, and the radial phase reported in Table 7 uses the adopted Galactocentric frame but no orbit integration and no potential at all. What the potential supplies is the depth and timing of that first approach. Beyond it the information decays quickly. The median radial period of the catalogue is 91 Myr (16th–84th percentile 80–109 Myr), and in the default barred potential Agama's normalised finite-time Lyapunov indicator has a median of 0.45 over the sampled chaos subset (96 stars; Appendix A), so neighbouring trajectories separate by a factor e in roughly two radial periods—of order 200 Myr, or about two pericentric passages. We therefore treat the first inward excursion as the interval over which a per-star trajectory is meaningfully constrained, and read everything beyond it, including the orbital history implied by integrating backwards, as a statistical property of the ensemble in the adopted potential rather than as the past of any individual star. This is a limitation of the data as much as of the method: a genuinely predictive long-baseline integration would require a time-dependent Milky Way model, including the bar's history and the response of the disc, which present observations do not constrain.

A present-day clustering null test (a DBSCAN sensitivity grid over $\varepsilon = \{5, 10, 15, 25, 50\}$ pc and $\text{min_samples} = \{3, 4, 5, 7, 10\}$; Ester et al. 1996; Schubert et al. 2017) returns zero compact groups for both Tier A+B and Tier A+B+C, so the catalogue is not explained by present-day bound clusters or compact associations; this does not rule out common accretion-related origins or action-space substructure, which need not remain compact in present-day position. The grid and nearest-pair statistics are in Appendix D (Table D.1). This present-day compactness check is deliberately narrower than the standard integral-of-motion and action-space clustering programme for identifying accreted debris (Helmi & de Zeeuw 2000; Myeong et al. 2018a,b; Koppelman et al. 2019; Lövdal et al. 2022; Ruiz-Lara et al. 2022; Dodd et al. 2023). Density-based searches, including DBSCAN-like searches in integral-of-motion space (Borsato et al. 2020), ask whether stars group into common-origin debris in approximately conserved coordinates. Our null test instead asks whether the low- V_{grf} catalogue is simply one or more compact present-day spatial aggregates; it is not a progenitor-assignment or full action-space substructure analysis.

The spectroscopic cross-match sharpens the physical picture while stopping short of an origin claim. The alpha-classified subset contains only 117 Tier A+B+C stars, so population-region fractions are quoted only inside that subset. The sample overlaps phase-space regions often discussed in the Gaia-era accretion literature (Belokurov et al. 2018; Helmi et al. 2018; Belokurov et al. 2023), and the high-eccentricity chemistry context has been studied with much larger spectroscopic samples (Lee et al. 2023). This paper is not a membership or merger-decomposition analysis. Because the sample is selected on instantaneous state, not on origin, chemistry, action space, or a fully forward-modelled selection function, we do not attempt progenitor assignment here.

As a selection-function scale diagnostic, we compute $V_{\text{eff}} = \sum_i 1/p_{\text{sel},i}$ over stars with valid GaiaUnlimited cells. For Tier A+B, excluding cells that sit exactly at the GaiaUnlimited prior mean gives $V_{\text{eff}} = 423$; including those cells at the prior gives $V_{\text{eff}} = 885$, a factor of 2.09 larger. For Tier A+B+C the corresponding values are $V_{\text{eff}} = 1,512$ and 3,406, a factor of 2.25. The difference is therefore best read as a

selection-function fallback systematic, not as a volume-complete population normalization.

The prior-dominated (low-parent-count, $n < 10$) fraction can be quantified directly from the underlying GaiaUnlimited (G , $BP-RP$, HEALPix) grid. For Tier A+B, 83.2% of valid stars sit in cells with parent count $n < 10$ and contribute 88.7% of $\sum 1/S_{\text{RVS}}$ (Table 16; Fig. 11); restricting to the inner-Galaxy quadrant $l \in [330^\circ, 30^\circ]$ where the catalogue concentrates lowers this to 69.7% of stars and 76.8% of $\sum 1/S_{\text{RVS}}$ – still majority prior-dominated. The choice of selection-function estimator is a smaller effect. Combining the Cantat-Gaudin et al. (2023) DR3 parent-completeness map with the Castro-Ginard et al. (2023) RVS subsample SF leaves $\sum 1/S_{\text{RVS}}$ for Tier A+B unchanged at 885 to within 0.1% because the DR3 parent completeness saturates at the catalogue’s $G_{\text{RVS}} < 14$ magnitude range. Substituting the older EDR3 RVS SF of Everall & Boubert (2022) yields a nominally $\sim 18\times$ larger $\sum 1/S$, but that estimator targets a smaller, brighter RVS subsample than DR3 RVS and is not a like-for-like replacement. We therefore retain the prior-dominance audit as the operative SF systematic for the catalogue.

The principal open questions are interpretive rather than catalogue-level. Spectroscopic coverage is incomplete. Very small inferred cylindrical pericentres and long-apocentre inner-reach counts are model-sensitive. The sample is not volume-complete, and local halo samples cannot in general be extrapolated to the global halo without strong modelling assumptions (Sharpe et al. 2024). These limitations set the boundary of the present paper’s claims.

6. Conclusions

We present a tiered, probabilistic Gaia DR3 catalogue of stars with very low Galactic rest-frame speeds: a primary Tier A+B catalogue of 276 stars ($P(V_{\text{grf}} < 25 \text{ km s}^{-1}) > 0.84$), a high-confidence Tier A core of 173 ($P > 0.95$), and a high-completeness Tier A+B+C extension of 621 ($P > 0.50$). Membership probabilities are corrected for the population prior; the uncorrected forward scores select 289, 541, and 1,952 stars at the same thresholds and are also released. Each source carries a Monte Carlo membership probability, so the catalogue is a probability-scored sample rather than a hard-threshold list.

In the axisymmetrized Hunter+2024 potential the sample is strongly radial with compact cylindrical pericentres (Monte Carlo posterior-median eccentricity 0.949, $R_{\text{peri}} = 154 \text{ pc}$, $R_{\text{apo}} = 7.35 \text{ kpc}$). This radial character is largely a corollary of the selection—a near-zero instantaneous speed picks out stars near an orbital turning point—and is robust across the halo-mass, barred, and second-potential (McMillan+2017, Portail+2017) tests of Appendix A. The matched-control comparison is a null check, not a calibrated relation: against observability-matched samples at higher V_{grf} the slow catalogue is distinctly more eccentric and compact in cylindrical pericentre, and the across-band gradient runs in the expected kinematic sense, but the controls are a deliberately broad, selection-function-limited comparison over four coarse velocity bands. We therefore conclude that the slow state is not a generic property of matched 6D stars, and we do not claim a quantitative speed–eccentricity law—that would require a substantially larger, selection-corrected control sample.

The exact cylindrical inner-Galaxy reach is model-dependent and we treat it as such. The long-apocentre inner-reach count rises from three in the static potential to 7–31 across the barred pattern-speed grid ($\Omega_p = 24, 28, 33, 37.5, 41 \text{ km s}^{-1} \text{ kpc}^{-1}$),

with an independent Portail+2017 M2M bar giving a consistent 15 (Table 21); the parsec-scale Sgr A* approach is weaker still—six soft candidates defined only by Monte Carlo posterior probability, none secure below 10 pc (Appendix B). Stäckel-fudge actions are reliable in J_R and L_z but not J_z for these plunging orbits, so the catalogue ships orbit-time-averaged actions and a per-star reliability flag alongside the single-evaluation values.

The orbit products are also bounded in time, and we would rather state this than leave it implied. What the data fix is the next excursion: a star observed at $V_{\text{grf}} < 25 \text{ km s}^{-1}$ is near an outer turning point and is falling inward next, and that follows from its measured velocity rather than from any potential. What the model supplies is how deep and how soon. With a median radial period of 91 Myr and a barred finite-time Lyapunov indicator implying e -folding trajectory separation in roughly two radial periods, per-star integrations stop being predictive after of order 200 Myr—about two pericentric passages. Statements beyond that horizon, in either time direction, should be read as ensemble properties in the adopted potential and not as the history or future of an individual star. Closing that gap needs a time-dependent Galaxy model, including the bar’s history and the disc’s response, which the present data do not constrain.

Completeness, not membership, is the dominant limitation: most of the catalogue occupies sparse, prior-dominated Gaia RVS selection-function cells, so inverse-selection sums are reported only as observability context and the catalogue is not volume-complete. Bar-induced chaos is present but bounded, and the Stäckel-fudge J_z caveat above means the orbital-shape statistics are descriptors rather than selection-defining. Chemistry is subset-qualified—only 117 Tier A+B+C stars carry the joint $[\text{Fe}/\text{H}] + \alpha$ abundances needed for literature-region comparison—and we make no whole-catalogue population assignment. A present-day DBSCAN search finds no bound clusters, consistent with a kinematically rather than spatially defined sample.

The natural next step is Gaia DR4: the same probabilistic low- V_{grf} selection on a longer baseline will test the stability of the present sample, retire the prior-dominated colour–magnitude cells that set the selection-function systematic, and enlarge the accessible low-speed catalogue.

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AI Assistance Disclosure. AI language-model tools were used during preparation of this manuscript for coding assistance, editorial revision, literature-navigation support, and critical feedback on draft text. These tools were not used as authors and were not treated as scientific authorities. In particular, no AI tool was used to generate, run, or independently validate the numerical analyses on which the catalogue rests: the Gaia DR3 sample construction, the Monte Carlo uncertainty propagation, the agama orbit integrations, the Galactocentric and threshold-membership calculations, the matched-control reweighting, the selection-function diagnostics, the statistical tests, and all figures and tables were produced and verified directly from the documented release-stage pipeline and re-executed by the author. All numerical results, validation steps, methodological decisions, interpretations, references, and final prose were reviewed and verified by the author, who takes sole responsibility for the scientific content.

Facilities.—Gaia, Sloan (APOGEE), AAT (HERMES/GALAH).

Software.—astropy (Astropy Collaboration et al. 2013, 2018, 2022), agama (Vasiliev 2019), GaiaUnlimited (Castro-Ginard et al. 2023), numpy, scipy, matplotlib, healpy, hdbscan, dustmaps.

Data Availability.—The observational source catalogues used in this work are publicly available from the respective survey archives (Gaia DR3 via the ESA Gaia Archive; APOGEE DR17 via the SDSS DR17 archive; GALAH DR3 via the GALAH Survey website; the Queiroz et al. 2023 StarHorse value-added catalogue via Vizier; and the Zhang et al. 2023

Gaia XP parameter catalogue via the GAVO Heidelberg TAP service as `xpparams.main`). The propagated candidate pool, the Tier A, Tier A+B, Tier A+B+C, and Tier D catalogues, the four matched-control band catalogues, the per-star Monte Carlo results, and the orbit-derived columns are archived in Zenodo at [doi:10.5281/zenodo.20116134](https://doi.org/10.5281/zenodo.20116134). The release includes the FITS catalogues, control catalogues, validation products, Gaia Archive selection query, pipeline scripts, potential configuration files, and environment metadata needed to reproduce the numerical products from the released intermediate tables.

Software Availability.—The review-stage analysis pipeline, including sample construction, control-band matching, orbit integration, Monte Carlo propagation, and the figure- and table-generation scripts, is available in a version-controlled GitHub repository at github.com/wodvik/gaia-slow-vgrf-catalogue. The repository also contains the review-stage catalogue products, column documentation, environment file, configuration file, Gaia Archive selection query, and the current manuscript draft.

Appendix A: Robustness and Sensitivity Battery

The principal dynamical claims are tested here against quality filtering, threshold variation, measurement uncertainty, distance estimator, Solar frame, halo mass, and barred-potential sensitivity.

Gold-subset dynamical robustness in the adopted model.

The Gold dynamical subset starts from the Tier A+B primary catalogue and applies the observational Gold cut before comparing point-estimate orbit summaries. The Gold cut barely changes the orbit distribution: median R_{peri} is 114.8 pc in the Gold subset versus 112.5 pc in the full Tier A+B catalogue, and all Gold stars have finite action solutions in the adopted static potential (Table A.1). The subset is intended for individual-star follow-up rather than for redefining the full catalogue.

Table A.1. Orbit-derived properties before and after the observational Gold cut. The Gold subset is defined in Table 11; quantities here are point-estimate outputs of the adopted static potential, including eccentricity, and are not used to define the subset. Eccentricity-fraction errors are $\text{Beta}(k + \frac{1}{2}, n - k + \frac{1}{2})$ Jeffreys 68% intervals (Cameron 2011).

Quantity	Tier A+B	Gold obs.
N stars	276	226
Median eccentricity	0.973	0.972
Fraction with $e > 0.8$ (%)	100.0	100.0
Fraction with $e > 0.95$ (%)	$80.4^{+2.3}_{-2.5}$	$79.2^{+2.6}_{-2.8}$
Fraction with $e > 0.99$ (%)	$17.8^{+2.4}_{-2.2}$	$17.7^{+2.7}_{-2.4}$
Median R_{peri} (pc)	114.0	114.8
Median R_{apo} (kpc)	8.24	8.18
Median z_{max} (kpc)	4.12	4.05
Finite action solutions	276	226

Threshold sensitivity within the current local sample. The 25 km s^{-1} threshold is a practical operating cut, not a special physical boundary. The candidate pool is therefore released with per-star $P(V_{\text{grf}} < 25 \text{ km s}^{-1})$ values and tier labels rather than only a hard threshold flag; alternative threshold cuts can be reconstructed directly from the machine-readable catalogue.

Table A.2. Threshold sensitivity. Counts at varying V_{grf} cuts and the corresponding orbital eccentricity distribution under the point-estimate Hunter+2024 orbit integration. The $V_{\text{grf}} < 25 \text{ km s}^{-1}$ threshold used for the primary catalogue is a practical operating cut, not a sharp physical boundary. Fraction errors are $\text{Beta}(k + \frac{1}{2}, n - k + \frac{1}{2})$ Jeffreys 68% intervals (Cameron 2011). Point-estimate fractions are systematically higher than the Monte Carlo posterior-median counterparts in Table 18 because of measurement-error convolution.

Cut (km s^{-1})	N_{point}	N_{ABC}	$\bar{e}$	$e > 0.95$ (%)
< 10	188	182	0.986	$100.0_{-0.5}^{+1.3}$
< 15	588	46	0.983	$97.8^{+1.3}_{-2.3}$
< 20	1,420	199	0.976	$85.4^{+2.3}_{-2.7}$
< 25	2,755	621	0.970	$74.9^{+1.7}_{-1.8}$

Velocity-threshold Monte Carlo propagation. Adaptive Monte Carlo propagation was run for all 20,829 near-threshold candidates. The baseline pass used 500 realisations per source,

with 5,000-realisation refinement in the transition region and 10,000-realisation refinement near the hard 25 km s^{-1} point-estimate boundary. The resulting tier labels, rather than a single point estimate, define the catalogue (Table A.3).

Table A.3. Adaptive Monte Carlo velocity-threshold propagation for the candidate pool. Each star is sampled from the Gaia DR3 parallax–proper-motion covariance, with sky position held fixed, an independent radial-velocity error, and the Bailer-Jones+2021 photogeometric distance posterior. Transition-band stars are refined to 5,000 realisations; near-threshold point estimates are refined to 10,000. The table reports the actual number of stars in each adaptive branch.

Quantity	Value
Candidate pool	20,829
Tier A+B+C after MC tiering	621
Pass-1 realisations / star	500
Transition-band refinements	1,623
Near-threshold ultra refinements	1,261
Refinement realisations / star	5,000
Ultra realisations / star	10,000
Minimum / maximum recovered $P(V_{\text{grf}} < 25 \text{ km s}^{-1})$	0.0 / 1.0

Halo-mass sensitivity on the full Tier A+B+C catalogue. A full-sample halo-mass sensitivity study was repeated for the Tier A+B+C catalogue, bracketing recent Milky Way mass estimates (e.g. Cautun et al. 2020; Wang et al. 2020). Varying the halo mass by $\pm 15\%$ changes the median point-estimate cylindrical pericentre from 116 pc in the matched 40,001-point default integration to 119 pc and 111 pc, respectively, and leaves the long-apocentre inner-reach count unchanged at three. The high-eccentricity result is therefore driven primarily by the observed phase-space configuration: all 621 Tier A+B+C stars have $e > 0.8$ in the default static potential.

Bar-perturbation sensitivity. Because many catalogue stars reach the inner Galaxy, we integrate a barred Hunter/Sormani variant (Sormani et al. 2022) over the pattern-speed grid $\Omega_p = 24, 28, 33, 37.5, 41 \text{ km s}^{-1} \text{ kpc}^{-1}$, bracketing the slower $\Omega_p \approx 24 \pm 3 \text{ km s}^{-1} \text{ kpc}^{-1}$ bar of Horta Darrington et al. (2025) and the commonly adopted $35\text{--}45 \text{ km s}^{-1} \text{ kpc}^{-1}$ range (Clarke et al. 2019; Zhang et al. 2024). Barred integration leaves the qualitative low- V_{grf} radiality intact but raises the long-apocentre inner-reach count from three (static) to 21 (barred default), spanning 7–31 across the grid, with the bar-angle and halo-flattening ranges given in Tables A.4 and A.6. The barred median cylindrical pericentre is itself sampling-sensitive (20 pc to 14 pc over 20,001–40,001 trajectory points), so we rely on threshold counts rather than barred medians for inner-reach comparisons and retain the barred integrations as systematic tests rather than catalogue values.

Because the bar can induce chaos on the R_{peri} scales sampled here (Dillamore et al. 2023, 2024; Woudenberg & Helmi 2025), we computed Agama’s normalized finite-time Lyapunov indicator λT_{orb} at $\Omega_p = 37.5 \text{ km s}^{-1} \text{ kpc}^{-1}$ for the three static-potential inner-reach candidates, the ten Sgr A* approacher seeds (Table B.1), and a random 100-star Tier A control sample. Bar-induced chaos is present but bounded: none of the three reach candidates, one of ten seeds, and four of 100 controls reach the order-unity strongly-chaotic regime, and the closest-approach seeds are not separated from the controls by the chaos indicator (Fig. A.1).

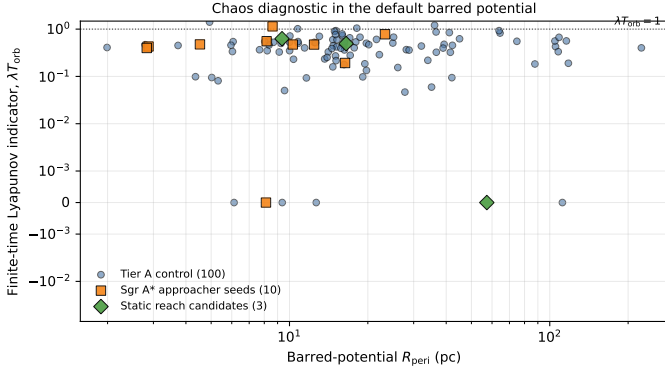


Fig. A.1. Finite-time Lyapunov indicator in the default barred Hunter/Sormani potential at $\Omega_p = 37.5 \text{ km s}^{-1} \text{ kpc}^{-1}$, plotted against the barred-potential cylindrical pericentre. The horizontal dotted line marks $\lambda T_{\text{orb}} = 1$, the order-unity level used by Agama as a practical strongly chaotic threshold.

As a second-potential check, re-integrating the catalogue in the made-to-measure Portail et al. (2017) bar—an independent disc and flattened Einasto dark halo, constructed via the Sormani et al. (2022) recipe at $\Omega_p = 37.5 \text{ km s}^{-1} \text{ kpc}^{-1}$ and a 25° bar angle—gives $N_{\text{reach}} = 15^{+5.0}_{-3.8}$ (Gehrels 1986 68%) and the same high-eccentricity character, overlapping the barred Hunter+2024 value $21^{+5.7}_{-4.5}$ within Gehrels 1σ (Table 21). The inner-reach result is therefore robust to swapping the disc/halo background at fixed bar parameterisation; the stricter 3D Sgr A* closest-approach probabilities are treated separately in Appendix B.

Radial-velocity uncertainty robustness. As an additional defensive cut, we required Gaia DR3 radial-velocity uncertainty $\sigma_{RV} < 2 \text{ km s}^{-1}$ on top of the Gold filter. This removes the noisier RVS solutions and retains 171 of the 420 Gold stars ($40.7^{+2.4}_{-2.4}\%$). The comparatively aggressive reduction reflects that DR3 RVS uncertainties are intrinsically larger than 2 km s^{-1} for many of the cool giants that dominate the sample at the typical distance, where the magnitude-dependent inflation of σ_{RV} (Katz et al. 2023) interacts with the $\sim 25 \text{ km s}^{-1}$ velocity scale of interest. Within the surviving subset the median eccentricity remains high (0.968 vs. 0.971 for the full Gold sample), the median V_{grf} is similar (16.4 km s^{-1} vs. 15.8 km s^{-1}), and 100% retain $e > 0.8$. The high-eccentricity claim survives this additional quality cut.

Solar-parameter sensitivity. Table A.5 brackets the Solar-parameter dependence: across the literature range of R_0 , V_c , and the LSR motion, tier counts on the legacy preselection subsample vary by $\sim 50\%$, while the ensemble orbital geometry is stable to $\sim 12\%$ in R_{peri} and $\sim 1\%$ in R_{apo} and eccentricity.

Appendix B: Inner-Galaxy Close Approaches

The most extreme central-passage claims require additional caution. The orbit integrations are fully three-dimensional; only the separately reported R_{peri} summary is cylindrical. In the point-estimate products, the closest static and barred trajectories reach within 3.1 and 4.3 pc of the Galactic centre, respectively. Dedicated 5,000-realisation Monte Carlo propagation of the closest static and barred seeds shows that no candidate has $P(r_{\text{sph,min}} < 10 \text{ pc}) > 0.5$, while six candidates have $P(r_{\text{sph,min}} < 100 \text{ pc}) > 0.5$. These probabilities use the spherical

Table A.4. Galactic-potential sensitivity for the catalogue. Values, including the eccentricity columns, use point-estimate integrations for all Tier A+B+C stars; all rows use the same 40,001-point, parabola-interpolated trajectory readout. N_{reach} counts stars with $R_{\text{peri}} < 2 \text{ kpc}$ and $R_{\text{apo}} > 15 \text{ kpc}$. Barred median cylindrical pericentres are resolution-limited readouts at this sampling; long-apocentre inner-reach and cylindrical spin-axis-reach counts are the stable barred diagnostics. The McMillan+2017 row uses the independently calibrated, Gaia-DR3-external McMillan (2017) potential and reproduces the static Hunter+2024 result.

Variant	$\bar{e}$	$e > 0.95$ (%)
Hunter+2024 static	0.964	66.7
McMillan+2017 static	0.961	62.8
Halo mass 0.85×	0.963	64.4
Halo mass 1.15×	0.965	68.3
Barred $\Omega_p = 24$	0.995	93.9
Barred $\Omega_p = 28$	0.996	92.7
Bar angle sweep	≈ 0.996	92.3–95.2
Halo flattening sweep	0.963–0.964	65.2–67.0
Portail+2017 M2M bar	0.996	92.7

Variant	R_{peri} (pc)	R_{apo} (kpc)	N_{reach}
Hunter+2024 static	116	7.36	3
McMillan+2017 static	123	7.31	3
Halo mass 0.85×	119	7.37	3
Halo mass 1.15×	111	7.33	3
Barred $\Omega_p = 24$	$\lesssim 14$	7.89	8
Barred $\Omega_p = 28$	$\lesssim 14$	7.92	7
Bar angle sweep	$\lesssim 13$ – $\lesssim 14$	8.24–8.39	18–28
Halo flattening sweep	114–118	7.34–7.37	3–3
Portail+2017 M2M bar	$\lesssim 14$	8.38	15

Galactocentric closest approach, $r_{\text{sph}} = \sqrt{x^2 + y^2 + z^2}$, and are Monte Carlo posterior probability counts, not the point-estimate cylindrical R_{peri} counts above. Since $r_{\text{sph}} \geq \sqrt{x^2 + y^2}$, the Sgr A* set is a stricter subset of the cylindrical spin-axis-reach set. We therefore describe these as candidate inner-Galaxy approachers, not as a confirmed population of Sgr A*-passing stars. A three-point quadratic interpolation of $r_{\text{sph}}^2(t)$ around each sampled trajectory minimum is included in the probabilities in Table B.1. Figure B.1 shows the corresponding cumulative probability curves for the 20 closest point-estimate seeds in the static and barred-default integrations.

Table A.5. Solar-parameter sensitivity of the Tier-A/B/C catalogue counts and point-estimate median orbital geometry. The listed variants span the literature-quoted range of R_0 , $z_\odot$, V_c , and the LSR motion (GRAVITY Collaboration 2022; Schönrich 2012; Reid & Brunthaler 2020). Tier counts vary by $\sim 50\%$ across the variants, but ensemble orbital geometry is stable to $\sim 12\%$ in R_{peri} and $\sim 1\%$ in R_{apo} and eccentricity. Tier counts in this table are computed on the legacy preselection subsample used for the solar-frame re-tiering, so the absolute counts differ from the full-catalogue primary count (Tier A/B/C increments 173/103/345; Table 1). The adopted catalogue default is the 229 km s^{-1} convention documented in Section 2.4; the 232 km s^{-1} row is retained only as a legacy working-convention sensitivity row. The stable geometry medians are the quantity of interest here.

Variant	R_0 (kpc)	V_c (km s^{-1})	N_A	N_B	N_C	Med. R_{peri} (pc)	Med. R_{apo} (kpc)
legacy 232	8.178	232.0	215	116	300	150	8.10
GRAVITY+22	8.275	229.0	234	132	355	154	8.14
LSR-6 (S12)	8.178	232.0	249	156	365	141	7.99
RB20	8.178	248.5	176	100	239	163	8.10

Table A.6. Sensitivity battery status for the Tier A+B+C catalogue ($N = 621$). The default static and barred point-estimate orbit products have been regenerated from the parent-complete candidate pool. All median- e entries are point-estimate orbit summaries. Halo-mass and bar-pattern-speed rows are point-estimate sensitivity runs using the same initial conditions; the distance-estimator row remains a non-orbit tiering stress test.

Axis	Variant	Med. R_{peri} (pc)	Med. R_{apo} (kpc)	Med. e	N_{reach}
Default static	Hunter+2024 axisymmetric	116	7.36	0.964	3
Default barred	Hunter/Sormani barred, $\Omega_p = 37.5$	≤ 14	8.32	0.996	21
Halo mass	$0.85 \times M_{\text{halo}}$	119	7.37	0.963	3
Halo mass	$1.15 \times M_{\text{halo}}$	111	7.33	0.965	3
Bar pattern speed	$\Omega_p = 33 \text{ km s}^{-1} \text{ kpc}^{-1}$	≤ 15	8.28	0.996	8
Bar pattern speed	$\Omega_p = 41 \text{ km s}^{-1} \text{ kpc}^{-1}$	≤ 13	7.97	0.996	31
Distance estimator	BJ/parallax coupling stress test	—	—	—	no tier-scale change

Notes. The default rows use the higher-resolution orbit products reported in the main orbit table. Halo-mass and bar-speed rows use a uniform 40,001-point, parabola-interpolated sensitivity integration. The barred median cylindrical-pericentre readouts continue to fall between 20,001 and 40,001 points, so they are quoted as resolution-limited diagnostics; the long-apocentre inner-reach counts and $R_{\text{peri}} < 100 \text{ pc}$ counts are the stable barred comparisons.

Table B.1. Candidate Sgr A* approachers from the Tier A+B+C catalogue. Candidate rows are seeded from the closest point-estimate static and barred integrations, then re-evaluated with 5,000-realisation Monte Carlo orbit propagation and three-point quadratic interpolation of $r_{\text{sph}}^2(t)$ around each sampled minimum.

Gaia DR3 source_id	Tier	Potential	$r_{\text{sph,min}}$	p16	p50	p84	$P(< 10 \text{ pc})$ / $P(< 100 \text{ pc})$
				(pc)			
132425700141447168	C	barred	26	43	72		0.023 / 0.976
5871390894093751936	B	static	16	40	78		0.072 / 0.951
4047819072313279616	C	barred	20	43	84		0.041 / 0.895
132425700141447168	C	static	17	48	96		0.069 / 0.867
4047405312345623296	C	barred	23	53	172		0.027 / 0.754
6761283447501038848	C	barred	27	70	193		0.023 / 0.643
4036336430101629568	C	static	18	110	337		0.079 / 0.475
4038063281786427520	C	static	31	117	312		0.037 / 0.463
4074796625207486080	C	static	83	147	361		0.008 / 0.252
6629652011339174912	C	barred	90	222	349		0.003 / 0.179

Notes. No catalogue candidate satisfies $P(r_{\text{sph},\text{min}} < 10 \text{ pc}) > 0.5$. Six candidates satisfy the softer $P(r_{\text{sph},\text{min}} < 100 \text{ pc}) > 0.5$ criterion. These are spherical Galactocentric closest-approach probabilities from the Monte Carlo propagation and are distinct from the point-estimate cylindrical $R_{\text{peri}} < 100 \text{ pc}$ spin-axis-reach counts in Table 21. The closest point-estimate barred candidate is 6761283447501038848 with 4.3 pc. A source may appear twice when it is selected independently by the static and barred seed lists.

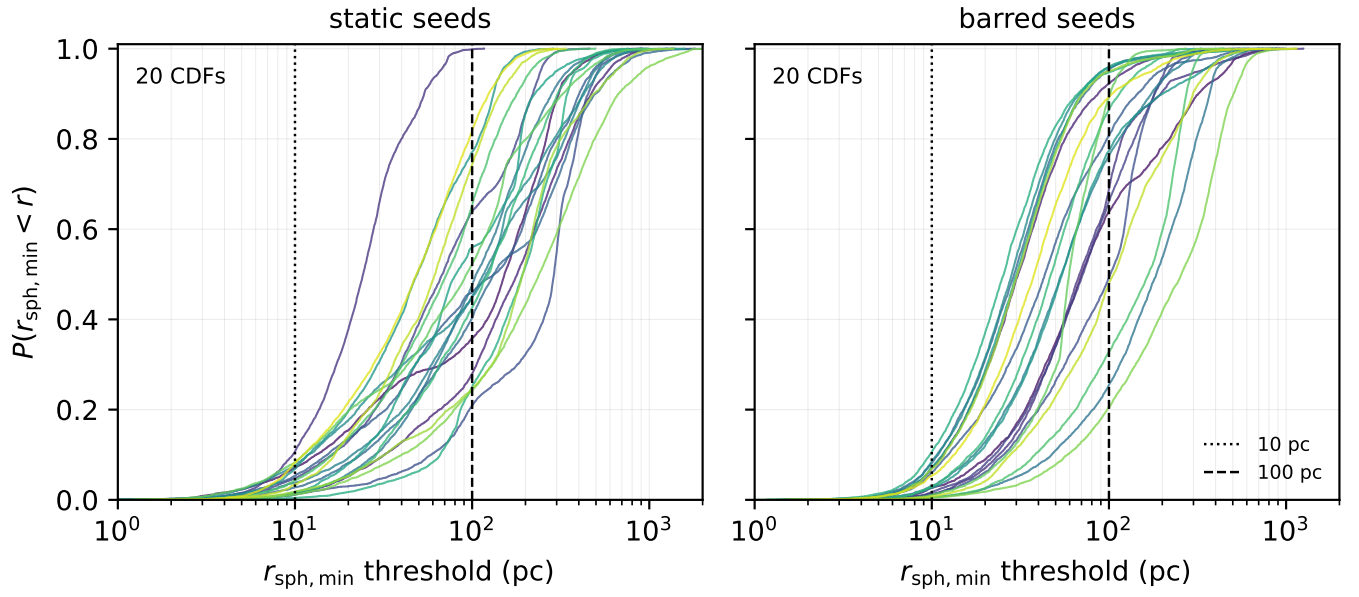


Fig. B.1. Monte Carlo cumulative probabilities for the 20 closest Sgr A* approachers in the static Hunter+2024 integration (left) and the barred-default integration (right). Each curve uses 5,000 propagated realisations and the same three-point interpolation of the sampled closest approach used in Table B.1. The vertical reference lines mark 10 and 100 pc.

Appendix C: Action-Space and Bar Resonances

The same L_z axis is the natural place to read off where the catalogue’s angular-momentum distribution sits relative to the bar resonances. Figure C.1 plots the Tier A+B+C catalogue in L_z – J_R Stäckel-fudge action space and overlays the inner Lindblad, corotation, and outer Lindblad loci derived from the axisymmetrised Hunter+2024 potential for the full pattern-speed grid $\Omega_p \in \{24, 28, 33, 37.5, 41\} \text{ km s}^{-1} \text{ kpc}^{-1}$. The Tier A points concentrate near $L_z \approx 0$ at large J_R , intersecting the ILR loci of several pattern speeds rather than piling up at any single corotation or OLR; this is the same low-angular-momentum, high-radial-action signature expected of compact-pericentre orbits, and it is consistent with the matched-control OLR-fraction check in Section 4.2.

We stress that this overlay is qualitative. The Stäckel-fudge actions on which it is built carry a large vertical-action uncertainty for these highly eccentric, inner-plunging orbits—the median absolute fractional J_z difference between catalogue and orbit-time-averaged actions is 85% in Tier A and 62% over the full Tier A+B+C sample, against $\lesssim 10\%$ for J_R and L_z (Appendix A)—so the figure should be read as locating the catalogue’s low- L_z , high- J_R concentration relative to the resonance loci, not as a quantitative resonance-occupancy classification. The weighted matched-control comparison does not favour a simple outer-Lindblad pile-up explanation.

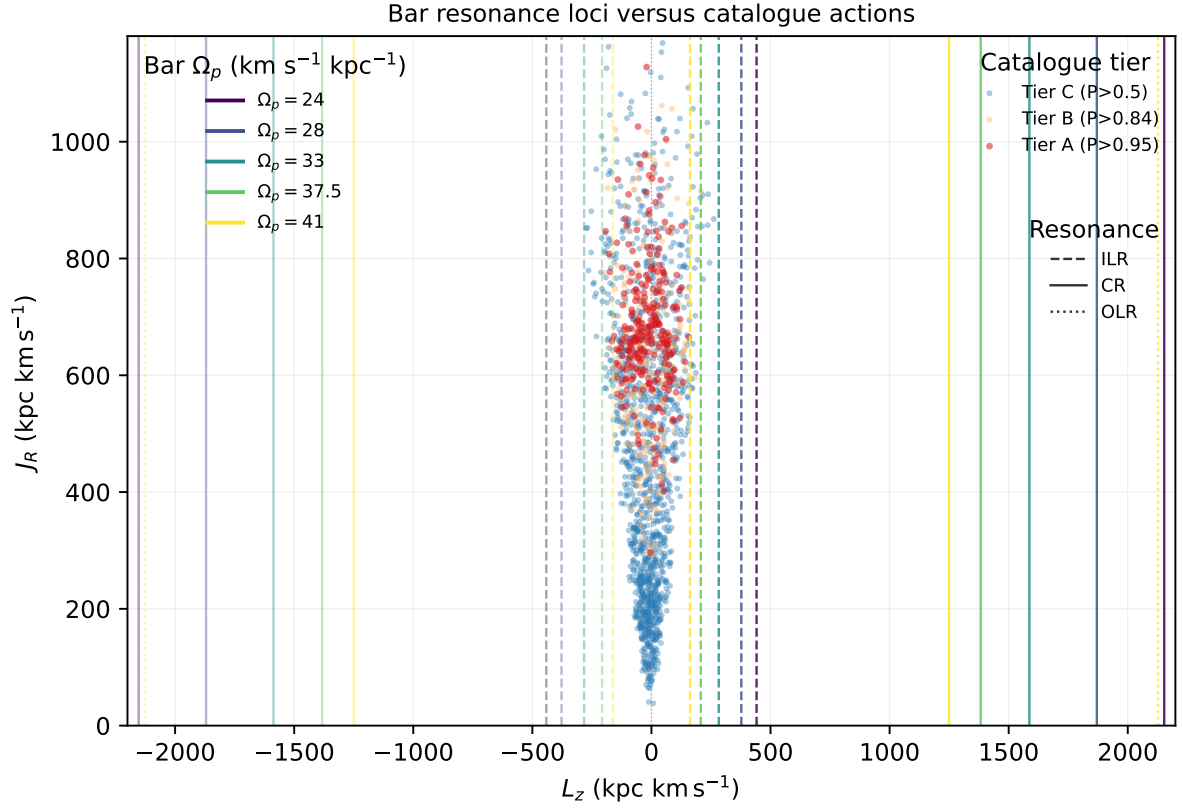


Fig. C.1. Bar resonance loci versus catalogue actions. Vertical lines mark the values of L_z for circular orbits at the inner Lindblad (dashed), corotation (solid), and outer Lindblad (dotted) radii in the axisymmetrised Hunter+2024 potential at each of the five bar pattern speeds in the extended grid (colour-coded). Loci at negative L_z mirror the retrograde-orbit case. Catalogue points are colour-coded by tier and use the Stäckelfudge (L_z , J_R) values reported per star; Tier A points cluster near $L_z \approx 0$ at large J_R .

Appendix D: Present-Day Clustering Null Test

As a final present-day compactness check, we searched the slow catalogue for open-cluster- or association-like aggregates in 3D Galactocentric position (Table D.1). The Tier A+B sample contains one pair below 25 pc, with a closest-pair separation of 20.2 pc. The broader Tier A+B+C sample contains two close pairs below 25 pc, but a DBSCAN sensitivity grid (Ester et al. 1996; Schubert et al. 2017) over $\varepsilon = \{5, 10, 15, 25, 50\}$ pc and $\text{min_samples} = \{3, 4, 5, 7, 10\}$ returns zero compact groups for both Tier A+B and Tier A+B+C. Thus the catalogue is not explained by one or more present-day bound clusters or compact associations. This does not rule out common accretion-related origins or action-space substructure, which need not remain compact in present-day position and is more naturally addressed with the integral-of-motion and action-space methods developed for halo debris searches (Helmi & de Zeeuw 2000; Borsato et al. 2020; Lövdal et al. 2022; Dodd et al. 2023).

Table D.1. DBSCAN present-day compact-group sensitivity grid for the low- V_{grf} catalogue. Entries give the number of compact groups found in 3D Galactocentric position space for Tier A+B / Tier A+B+C.

ε (pc)	min_samples				
	3	4	5	7	10
5	0/0	0/0	0/0	0/0	0/0
10	0/0	0/0	0/0	0/0	0/0
15	0/0	0/0	0/0	0/0	0/0
25	0/0	0/0	0/0	0/0	0/0
50	0/0	0/0	0/0	0/0	0/0

Note. The search uses the Ester et al. (1996) DBSCAN algorithm in present-day 3D Galactocentric position space, with ε varied from 5 to 50 pc and min_samples from 3 to 10. The paired entries are Tier A+B / Tier A+B+C group counts. The closest Tier A+B pair is 20.2 pc, while Tier A+B+C contains two pairs below 25 pc but no three-member group at any grid point. This grid follows the parameter-sensitivity guidance of Schubert et al. (2017).

Appendix E: Distance-Covariance Tier Confusion Matrix

The tier-label confusion matrix under the maximal parallax-distance copula stress test of Section 2.3 is given below.

Table E.1. Tier-label confusion matrix for the full tier-sensitive band ($N = 3,342$ sources with adopted $P \geq 0.3$) under a maximal parallax-distance Gaussian-copula coupling, relative to an independent-distance recomputation at the same realisations. Rows are the independent-draw tier, columns the coupled-draw tier; off-diagonal entries are tier switches. Only 114 of 3,342 sources switch. Relative to the adopted primary catalogue, 276 of 276 Tier A+B members remain above $P = 0.84$ under the coupled draw (median $|\Delta P| = 0.0010$, max 0.0230).

indep.\copula	A	B	C	<C
A	287	7	0	0
B	4	235	8	0
C	0	13	1355	40
<C	0	0	42	1351

Appendix F: External Low-Azimuthal-Velocity Comparison

Filion et al. (2025) provide the closest published Gaia-era comparison sample selected by low Galactocentric azimuthal velocity rather than by low total Galactic rest-frame speed. Resolving their supplementary APOGEE identifiers to Gaia DR3 source identifiers gives 69 unique Gaia sources, all satisfying the present Gaia-RVS parent-source contract. Under the present low- V_{grf} diagnostic tiering, however, all 69 are Tier X: three enter the 20,829-source Monte Carlo candidate pool, but none enter Tier A, Tier A+B, or Tier A+B+C. Across the full Filion set, the same Gaia RVS quality audit classifies 66 sources as ok and three as poor; three have RV-variability flags and two are matched in the Gaia DR3 non-single-star two-body-orbit table. The three candidate-pool overlaps themselves are RVS-quality ok and NSS-clean, although one has RUWE just above 1.4. Thus the comparison separates two definitions of “slow”: Filion et al. select low azimuthal rotation with disc-like chemistry, whereas the present catalogue selects low total V_{grf} independent of chemistry and applies the same probability and quality gates to every source.

Table F.1. Filion et al. (2025) stars entering the present Monte Carlo candidate pool. V_{ϕ}^{F25} is an approximate reproduction of the Filion et al. Galactocentric azimuthal-velocity convention using APOGEE DR17 line-of-sight velocities and Bailer-Jones et al. (2021) photogeometric distances. The remaining columns are from the present Gaia DR3 RVS low- V_{grf} release products. None of the three sources is matched in the Gaia DR3 non-single-star two-body-orbit table. All three are classified as Tier X and are therefore excluded from the Tier A+B+C science catalogue.

Gaia DR3 source_id	APOGEE ID	V_{ϕ}^{F25} (km s ⁻¹)	V_{grf} (km s ⁻¹)	$P_{<25}$	σ_{RV} (km s ⁻¹)	N_{tr}	$p_{\chi^2, \text{RV}}$	RUWE	RV class
3685921348377044480	2M13041724-0134399	-37.5	27.5	0.364	1.17	18	—	0.95	ok
1594059695819370624	2M14575158+5300140	15.2	40.0	0.090	1.14	26	0.640	1.17	ok
1288956935783694464	2M15022633+3248296	-24.8	47.6	0.000	0.19	33	0.924	1.41	ok

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